ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

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PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

Exercise

Kinetic Theory of Gases & Thermodynamics

(English Medium)

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

KTG

1. Find the approx. number of molecules contained in a vessel of volume 7 litres at 0°C at 1.3×10^5 pascal

(A) 2.4×10^{23} (B) 3×10^{23} (C) 6×10^{23} (D) 4.8×10^{23}

KT0029

2. An ideal gas mixture filled inside a balloon expands according to the relation $PV^{2/3}$ = constant. The temperature inside the balloon is

(A) increasing (B) decreasing (C) constant (D) can't be said

KT0030

3. At a temperature T K, the pressure of 4.0g argon in a bulb is p. The bulb is put in a bath having temperature higher by 50K than the first one. 0.8g of argon gas had to be removed to maintained original pressure. The temperature T is equal to

(A) 510 K (B) 200 K (C) 100 K (D) 73 K

KT0031

4. 28 gm of N_2 gas is contained in a flask at a pressure of 10 atm and at a temperature of 57°. It is found that due to leakage in the flask, the pressure is reduced to half and the temperature reduced to 27°C. The quantity of N_2 gas that leaked out is :-

(A) 11/20 gm (B) 20/11 gm (C) 5/63 gm (D) 63/5 gm

KT0032

5. A container X has volume double that of container Y and both are connected by a thin tube. Both contains same ideal gas. The temperature of X is 200K and that of Y is 400K. If mass of gas in X is m then in Y it will be:

(A) m/8 (B) m/6 (C) m/4 (D) m/2

KT0033

6. When 2 gms of a gas are introduced into an evacuated flask kept at 25^0C the pressure is found to be one atmosphere. If 3 gms of another gas added to the same flask the pressure becomes 1.5 atmospheres. The ratio of the molecular weights of these gases will be :-

(A) 1 : 3 (B) 3 : 1 (C) 2 : 3 (D) 3 : 2

KT0034

7. A rigid tank contains 35 kg of nitrogen at 6 atm. Sufficient quantity of oxygen is supplied to increase the pressure to 9 atm, while the temperature remains constant. Amount of oxygen supplied to the tank is :

(A) 5 kg (B) 10 kg (C) 20 kg (D) 40 kg

KT0035

8. During an experiment an ideal gas obeys an addition equation of state P^2V = constant. The initial temperature and volume of gas are T and V respectively. When it expands to volume 2V, then its temperature will be:

(A) T (B) $\sqrt{2}$ T (C) 2 T (D) $2\sqrt{2}$ T

KT0036

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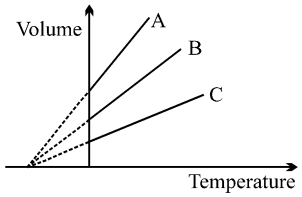
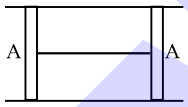
9. A vessel contains 1 mole of O_2 gas (molar mass 32) at a temperature T. The pressure of the gas is P. An identical vessel containing one mole of He gas (molar mass 4) at a temperature 2T has a pressure of
(A) P/8 (B) P (C) 2P (D) 8P **KT0037**

10. The expansion of an ideal gas of mass m at a constant pressure P is given by the straight line B. Then the expansion of the same ideal gas of mass 2 m at a pressure 2P is given by the straight line



(A) C (B) A (C) B (D) none **KT0038**

11. A cylindrical tube of cross-sectional area A has two air tight frictionless pistons at its two ends. The pistons are tied with a straight piece of metallic wire. The tube contains a gas at atmospheric pressure P_0 and temperature T_0. If temperature of the gas is doubled then the tension in the wire is

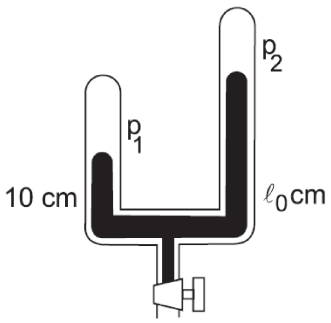


(A) 4 P_0 A (B) $P_0A/2$ (C) P_0 A (D) 2 P_0 A **KT0039**

12. In a constant volume gas thermometer, the pressure of the working gas is measured by the difference in the levels of mercury in the two arms of a U-tube connected the the gas at one end. When the bulb is placed at the room temperature 27.0°C, the mercury column in the arm open to atmosphere stands 5.00 cm above the level of mercury in the other arm. When the bulb is placed in a hot liquid, the difference of mercury levels becomes 45.0 cm. Calculate the temperature of the liquid. (atmospheric pressure = 75.0 cm of mercury.)
(A) 450°C (B) 177°C (C) 40.5°C (D) 243 °C **KT0281**

13. The pressure of the gas in a constant volume gas thermometer are 80 cm, 90 cm and 100 cm of mercury at the ice point, the steam point and in a heated wax bath respectively. Find the temperature of the wax bath.
(A) 150 °C (B) 200°C (C) 250 °C (D) 300°C **KT0282**

14. A mercury manometer (figure 24 W1) consists of two unequal arms of equal cross-section 1 cm^3 and lengths 100 cm and 50 cm . The two open ends are sealed with air in the tube at a pressure of 80 cm of mercury. Some amount of mercury is now introduced in the manometer through the stopcock connected to it. If mercury rises in the shorter tube to a length 10 cm in steady state, find the length of the mercury column risen in the longer tube.



(A) 15 cm (B) 20 cm (C) 15.5 cm (D) 25 cm **KT0283**

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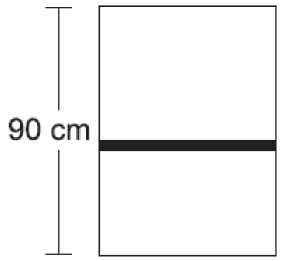
15. A barometer reads 75 cm of mercury. When 2.0 cm^3 of air at atmospheric pressure is introduced into the space above the mercury level, the volume of this space becomes 50 cm^3. Find the length by which the mercury column descends.

(A) 36 cm (B) 72 cm (C) 3 cm (D) 50 cm **KT0284**

16. Figure shows a vertical cylindrical vessel separated in two parts by a frictionless piston free to move along the length of the vessel. The length of the cylinder is 90 cm and the piston divides the cylinder in the ratio of 5 : 4. Each of the two parts of the vessel contains 0.1 mole of an ideal gas. The temperature of the gas is 300 K in each part. Calculate the mass of the piston.



(A) 12.7 kg (B) 16.9 kg (C) 21 kg (D) 18.2 kg **KT0285**

17. Figure shows a cylindrical tube of volume V_0 divided in two parts by a frictionless separator. The walls of the tube are adiabatic but the separator is conducting. Ideal gases are filled in the two parts. When the separator is kept in the middle, the pressures are p_1 and p_2 in the left part and the right part respectively. The separator is slowly slid and is released at a position where it can stay in equilibrium. Find the volumes of the two parts.

(A) $V_1 = \frac{|p_1 - p_2|V_0}{p_1 + p_2}$ and $V_2 = \frac{p_1 + p_2}{|p_1 + p_2|}V_0$ (B) $V_1 = \frac{p_1 + p_2}{|p_1 - p_2|}V_0$ and $V_2 = \frac{|p_1 - p_2|}{p_1 + p_2}V_0$

(C) $V_1 = \frac{p_1 V_0}{p_1 + p_2}$ and $V_2 = \frac{p_2 V_0}{p_1 + p_2}$ (D) $V_1 = \frac{p_2 V_0}{p_1 + p_2}$ and $V_2 = \frac{p_1 V_0}{p_1 + p_2}$

KT0286

18. Equal masses of air are sealed in two vessels, one of volume V_0 and the other of volume $2V_0$. If the first vessel is maintained at a temperature 300 K and the other at 600 K, find the ratio of the pressures in the two vessels.

(A) 1 : 2 (B) 2 : 1 (C) 1 : 4 (D) 1 : 1 **KT0287**

19. An electric bulb of volume 250 cc was sealed during manufacturing at a pressure of 10^{-3} mm of mercury at 27°C. Compute the number of air molecules contained in the bulb. Avogadro constant = 6×10^{23} per mol, density of mercury = 13600 kg/m^3 and g = 10 m/s^2.

(A) 6×10^{15} (B) 9×10^{15} (C) 12×10^{15} (D) 8.0×10^{15} **KT0288**

20. A gas cylinder has walls that can bear a maximum pressure of 1.0×10^6 Pa. It contains a gas at 8.0×10^5 Pa and 300K. The cylinder is steadily heated. Neglecting any change in the volume, calculate the temperature at which the cylinder will break.

(A)400 K (B) 225 K (C) 350 K (D) 375 K **KT0289**

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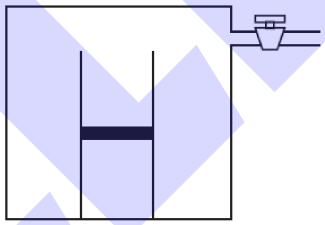
21. An air bubble of radius 2.0 mm is formed at the bottom of a 3.3 m deep river. Calculate the radius of the bubble as it comes to the surface. Atmospheric pressure = 1.0×10^5 Pa and density of water = 1000 kg/m^3.
(A) 1.5 mm (B) 2.4 mm (C) 2.2 mm (D) 2.1 mm **KT0290**

22. A uniform tube closed at one end, contains a pallet of mercury 10 cm long. When the tube is kept vertically with the closed end upward, the length of the air column trapped is 20 cm. Find the length of the air column trapped when the tube is inverted so that the closed end goes down. Atmospheric pressure = 75 cm of mercury.
(A) 10 cm (B) 12 cm (C) 15 cm (D) 18 cm **KT0291**

23. An ideal gas is trapped between a mercury column and the closed end of a narrow vertical tube of uniform base containing the column. The upper end of the tube is open to the atmosphere. The atmospheric pressure equals 76 cm of mercury. The lengths of the mercury column and the trapped air column are 20 cm and 43 cm respectively. What will be the length of the air column when the tube is tilted slowly in a vertical plane through an angle of 60^0? Assume the temperature to remain constant.
(A) 48 cm (B) 45 cm (C) 40 cm (D) 50 cm **KT0292**

24. An ideal gas is kept in a long cylindrical vessel fitted with a frictionless piston of cross-sectional area 10 cm^2 and weight 1 kg (figure). The vessel itself is kept in a big chamber containing air at atmospheric pressure 100 kPa. The length of the gas column is 20 cm. If the chamber is now completely evacuated by an exhaust pump, what will be the length of the gas column ? Assume the temperature to remain constant throughout the process.



(A) 2.0 m (B) 2.2 m (C) 2.5 m (D) 2.7 m **KT0293**

25. One mole of an ideal gas at STP is heated in an insulated closed container until the average speed of its molecules is doubled. Its pressure would therefore increase by factor.
(A) 1.5 (B) $\sqrt{2}$ (C) 2 (D) 4 **KT0041**

26. Three particles have speeds of 2u, 10u and 11u. Which of the following statements is correct?
(A) The r.m.s. speed exceeds the mean speed by about u.
(B) The mean speed exceeds the r.m.s. speed by about u.
(C) The r.m.s. speed equals the mean speed.
(D) The r.m.s. speed exceeds the mean speed by more than 2u. **KT0042**

27. The root mean square velocity of all molecules in a mixture containing CO_2 and N_2 in the mole ratio 1 : 7 is found to be x times the root mean square velocity of CO_2 molecules at the same temperature. The value of x is :
(A) $\sqrt{\frac{4}{5}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{\frac{2}{3}}$ (D) None **KT0294**

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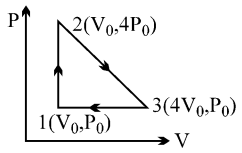
28. One mole of an ideal diatomic gas is taken through the cycle as shown in the figure.
$1 \rightarrow 2$: isochoric process
$2 \rightarrow 3$: straight line on P-V diagram
$3 \rightarrow 1$: isobaric process
The average molecular speed of the gas in the states 1, 2 and 3 are in the ratio



(A) 1 : 2 : 2 (B) $1 : \sqrt{2} : \sqrt{2}$ (C) 1 : 1 : 1 (D) 1 : 2 : 4 **KT0040**

29. Consider a sample of oxygen at 300K. Find the average time taken by a molecule to travel a distance equal to the diameter of the earth.
(A) 4 hours (B) 6 hours (C) 8 hour (D) 10 hours **KT0295**

30. Consider a gas for which diameter of molecules is σ, the gas is at a pressure P and temperature T and N_a = avagodro's number. The mean free path along x axis is :-
(A) $\frac{RT}{\sqrt{2}PN_a\pi\sigma^2}$ (B) $\frac{RT}{\sqrt{3}PN_a\pi\sigma^2}$ (C) $\frac{RT}{\sqrt{6}PN_a\pi\sigma^2}$ (D) $\frac{RT}{PN_a\pi\sigma^2}$
KT0296

31. A gas undergoes isobaric heating. The mean free path of the gas depends on temperature as $\lambda \propto T^a$. The number of collision that gas molecules undergo/sec depends on temperature as $\frac{dN}{dt} \propto T^b$. a and b are :
(A) $a = 0,\ b = \frac{1}{2}$ (B) $a = 1,\ b = \frac{1}{2}$ (C) $a = 1,\ b = -\frac{1}{2}$ (D) $a = -\frac{1}{2}, b = +\frac{1}{2}$
KT0297

32. Find the ratio of the mean speed of hydrogen molecules to the mean speed of nitrogen molecules in a sample containing a mixture of the two gases.
(A) 2.75 (B) 3.74 (C) 4.25 (D) 3.50 **KT0298**

33. Figure shows a vessel partitioned by a fixed diathermic separator . Different ideal gases are filled in the two parts. The rms speed of the molecules in the left part equals the mean speed of the molecules in the right part. Calculate the ratio of the mass of a molecule in the left part of the mass of a molecule in the right part.

(A) 0.82 (B) 1 (C) 1.18 (D) 1.4 **KT0299**

34. A gas mixture consists of 2 moles of oxygen and 4 moles of argon at temperature T. Neglecting all vibrational modes, the total internal energy of the system is :-
(A) 4 RT (B) 15 RT (C) 9 RT (D) 11 RT
KT0044

35. 0.040 g of He is kept in a closed container initially at 100^0C. The container is now heated. Neglecting the expansion of the container, calculate the temperature at which the internal energy is increased by 12 J.
(A) 469°C (B) 250 °C (C) 220 °C (D) 196°C **KT0300**

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36. Two monoatomic ideal gas at temperature T_1 and T_2 are mixed. There is no loss of energy. If the masses of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively. The temperature of the mixture will be :

(A) $\frac{T_1+T_2}{n_1+n_2}$ (B) $\frac{T_1}{n_1}+\frac{T_2}{n_2}$ (C) $\frac{n_2T_1+n_1T_2}{n_1+n_2}$ (D) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$

KT0046

37. The molar heat capacity of oxygen gas at STP is nearly 2.5 R. As the temperature is increased, it gradually increases and approaches 3.5 R. The most appropriate reason for this behaviour is that at high temperatures

(A) oxygen does not behave as an ideal gas (B) oxygen molecules dissociate in atoms
(C) the molecules dissociate in atoms (D) molecular vibration gradually become effective

KT0301

38. A gas contains only rigid diatomic molecules at temperature T. If I is the moment of inertia of the molecule, then mean square root angular velocity of the molecule is:

(A) $\sqrt{kT/I}$ (B) $\sqrt{3kT/I}$ (C) $\sqrt{5kT/I}$ (D) $\sqrt{2kT/I}$ **KT0302**

First law of thermodynamics

39. One mole of an ideal gas is contained with in a cylinder by a frictionless piston and is initially at temperature T. The pressure of the gas is kept constant while it is heated and its volume doubles. If R is molar gas constant, the work done by the gas in increasing its volume is :

(A) RT *l*n2 (B) 1/2 RT (C) RT (D) 3/2 RT

KT0043

40. In thermodynamic process pressure of a fixed mass of gas is changed in such a manner that the gas releases 30 joule of heat and 18 joule of work was done on the gas. If the initial internal energy of the gas was 60 joule, then, the final internal energy will be :

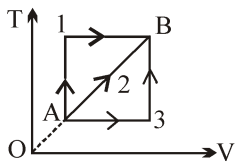
(A) 32 joule (B) 48 joule (C) 72 joule (D) 96 joule

KT0045

41. The first law of thermodynamics can be written as $\Delta U = \Delta Q + \Delta W$ for an ideal gas. Which of the following statements is correct?

(A) ΔU is always zero when no heat enters or leaves the gas
(B) ΔW is the work done by the gas in this written law.
(C) ΔU is zero when heat is supplied and the temperature stays constant
(D) $\Delta Q = -\Delta W$ when the temperature increases very slowly. **KT0054**

42. A given mass of a gas expands from a state A to the state B by three paths 1, 2 and 3 as shown in T-V indicator diagram. If W_1, W_2 and W_3 respectively be the work done by the gas along the three paths, then



(A) $W_1 > W_2 > W_3$ (B) $W_1 < W_2 < W_3$
(C) $W_1 = W_2 = W_3$ (D) $W_1 < W_2$, $W_1 > W_3$

KT0057

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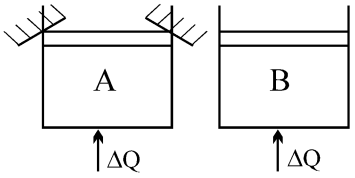
43. Two identical vessels A & B contain equal amount of ideal monoatomic gas. The piston of A is fixed but that of B is free. Same amount of heat is absorbed by A & B. If B's internal energy increases by 100 J the change in internal energy of A is :



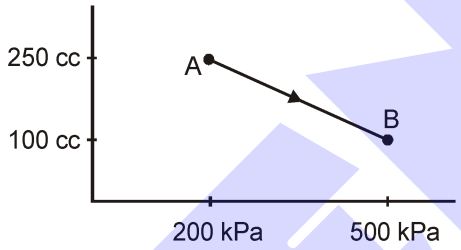
(A) 100 J (B) $\frac{500}{3}$ J

(C) 250 J (D) none of these **KT0063**

44. The internal energy of a monatomic ideal gas is 1.5 nRT. One mole of helium is kept in a cylinder of cross section 8.5 cm^2. The cylinder is closed by a light frictionless piston. The gas is heated slowly in a process during which a total of 42 J heat is given to the gas. If the temperature rises through 2°C, find the distance moved by the piston. Atmospheric pressure = 100 kPa.

(A) 10 cm (B) 15 cm (C) 20 cm (D) 25 cm **KT0303**

45. A gas is taken along the path AB as shown in figure. If 70 cal of heat is extracted from the gas in the process, calculate the change in the internal energy of the system.



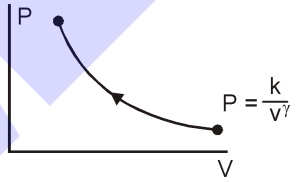
(A) –196 J (B) –221 J (C) – 241 J (D) –250 J **KT0304**

46. For a solid with a small expansion coefficient,

(A) Cp – Cv = R (B) Cp – Cv = R

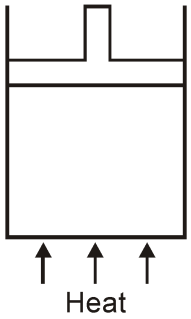
(C) Cp is slightly greater than Cv (D) Cp is slightly less than Cv **KT0305**

47. The molar heat capacity for the process shown in fig. is



(A) C = Cp (B) C = Cv (C) C > Cv (D) C = 0 **KT0306**

48. Figure shows a cylindrical container containing oxygen (γ = 1.4) and closed by a 50 kg frictionless piston. The area of cross-section is 100 cm^2, atmospheric pressure is 100 kPa and g is 10 m/s^2. Thy cylinder is slowly heated for some time. Find the amount of heat supplied to the gas if the position moves out through a distance of 20 cm.



(A) 980 J (B) 1050 J (C) 1220 J (D) 1280 J **KT0307**

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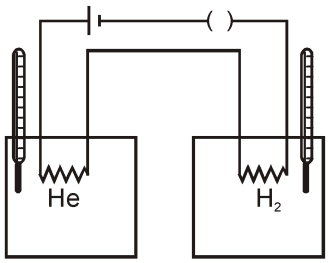
49. Figure shows two vessels with adiabatic walls, one containing 0.1 g of helium ($\gamma = 1.67$, M = 4 g/mol) and the other containing some amount of hydrogen ($\gamma = 1.4$, M = 2 g/mol). Initially, the temperatures of the two gases are equal The gases are electrically heated for some time during which equal amounts of heat are given to the two gases. It is found that the temperature rise through the same amount in the two vessels. Calculate the mass of hydrogen.



(A) 0.01 gm (B) 0.03 gm (C) 0.05 gm (D) 0.07 gm **KT0308**

50. One mole of an ideal monatomic gas is kept in a rigid vessel. The vessel is kept inside a steam chamber whose tempreature is 97°C. Initially, the temperature of the gas is 5.0°C. The walls of the vessel have an inner surface of area 800 cm^2 and thickness 1.0 cm. If the temperature of the gas increases to 9.0°C in 5.0 seconds, find the thermal conductivity of the material of the walls.

(A) 0.008 $Js^{-1}m^{-1}K^{-1}$ (B) 0.010 $Js^{-1}m^{-1}K^{-1}$ (C) 0.014 $Js^{-1}m^{-1}K^{-1}$ (D) 0.018 $Js^{-1}m^{-1}K^{-1}$

KT0309

51. A polyatomic gas with six degrees of freedom does 25J of work when it is expanded at constant pressure. The heat given to the gas is :-

(A) 100J (B) 150J (C) 200J (D) 250J

KT0049

52. A given quantity of an ideal gas is at pressure P and absolute temperature T. The isothermal bulk modulus of the gas is :

(A) 2P/3 (B) P (C) 3P/2 (D) 2P

KT0052

53. The adiabatic Bulk modulus of a diatomic gas at atmospheric pressure is

(A) 0 Nm^{-2} (B) 1 Nm^{-2} (C) 1.4×10^4 Nm^{-2} (D) 1.4×10^5 Nm^{-2}

KT0051

54. A reversible adiabatic path on a P-V diagram for an ideal gas passes through state A where $P = 0.7 \times 10^5$ N/m^{-2} and $V = 0.0049$ m^3. The ratio of specific heat of the gas is 1.4. The slope of path at A is :

(A) 2.0×10^7 Nm^{-5} (B) 1.0×10^7 Nm^{-5} (C) -2.0×10^7 Nm^{-5} (D) -1.0×10^7 Nm^{-5}

KT0050

55. The value of γ of linear arrangement of triatomic gas molecules is (Neglecting vibrational degree of freedom) :

(A) $\frac{9}{7}$ (B) $\frac{7}{5}$ (C) $\frac{5}{3}$ (D) $\frac{8}{6}$ **KT0310**

56. Two moles of monoatomic gas are mixed with 1 mole of a diatomic gas. The γ for the mixture is:

(A) 5/3 (B) 7/5 (C) 4/3 (D) 17/11 **KT0311**

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57. For an ideal diatomic gas in thermal equilibrium, the ratio of the molar heat capacity at constant volume at very high temperatures to that at room temperatures is equal to :
(A) 7/5 (B) 5/3 (C) 2 (D) 7/3 **KT0312**

58. Monoatomic, diatomic and triatomic gases whose initial volume and pressure are same, are compressed till their volume becomes half the initial volume.
(A) If the compression is adiabatic then monoatomic gas will have maximum final pressure.
(B) If the compression is adiabatic then triatomic gas will have maximum final pressure.
(C) If the compression is adiabatic then their final pressure will be same.
(D) If the compression is isothermal then their final pressure will be different. **KT0059**

59. An ideal gas expands from volume V_1 to V_2. This may be achieved by either of the three processes: isobaric, isothermal and adiabatic. Let ΔU be the change in internal energy of the gas, Q be the quantity of heat added to the system and W be the work done by the gas. Identify which of the following statements is false for ΔU?
(A) ΔU is least under adiabatic process.
(B) ΔU is greatest under adiabatic process.
(C) ΔU is greatest under the isobaric process.
(D) ΔU in isothermal process lies in-between the values obtained under isobaric and adiabatic processes.
KT0064

60. Three samples A, B, and C of the same gas ($\gamma = 1.5$) have equal volumes and temperatures. The volume of each sample is doubled, the process being isothermal for A, adiabatic for B and isobaric for C. If the final pressures are equal for the three samples, find the ratio of the initial pressures.
(A) 2 : 2 : 1 (B) $2\sqrt{2}$: 2 : 1 (C) 2 : $2\sqrt{2}$: 1 (D) 2 : 1 : $2\sqrt{2}$ **KT0313**

61. Figure shows a cylindrical tube with adiabatic walls and fitted with an adiabatic separator. The separator can be slid into the tube by an external mechanism. An ideal gas ($\gamma = 1.5$) is injected in the two sides at equal pressures and temperatures. The separator remains in equilibrium at the middle. It si now slid to a position where it divides the tube in the ratio 1 : 3. Find the ratio of the temperature in the two parts of the vessel.

(A) $\sqrt{2}$: 1 (B) $\sqrt{3}$: 1 (C) 1 : $\sqrt{2}$ (D) 1 : $\sqrt{3}$ **KT0314**

62. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio C_P/C_V for the gas is- **[AIEEE - 2003]**
(A) 4/3 (B) 2 (C) 5/3 (D) 3/2
KT0067

63. An ideal gas expands isothermally from a volume V_1 to V_2 and then compressed to original volume V_1 adiabatically. Initial pressure is P_1 and final pressure is P_3. The total work done is W. Then
[JEE' 2004 (Scr)]
(A) $P_3 > P_1$, $W > 0$ (B) $P_3 < P_1$, $W < 0$
(C) $P_3 > P_1$, $W < 0$ (D) $P_3 = P_1$, $W = 0$ **KT0047**

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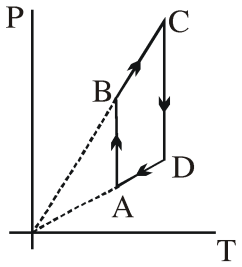
64. Pressure versus temperature graph of an ideal gas is shown in figure



(A) During the process AB work done by the gas is positive.

(B) During the process CD work done by the gas is negative.

(C) During the process BC internal energy of the gas is increasing.

(D) None **KT0048**

65. One mole of an ideal gas at temperature T_1 expends according to the law $\frac{P}{V^2}$ = a (constant). The work done by the gas till temperature of gas becomes T_2 is :

(A) $\frac{1}{2}R(T_2 - T_1)$ (B) $\frac{1}{3}R(T_2 - T_1)$ (C) $\frac{1}{4}R(T_2 - T_1)$ (D) $\frac{1}{5}R(T_2 - T_1)$

KT0053

66. 2 moles of a monoatomic gas are expanded to double its initial volume, through a process P/V = constant. If its initial temperature is 300 K, then which of the following is not true.

(A) $\Delta T = 900$ K (B) $\Delta Q = 3200$ R (C) $\Delta Q = 3600$ R (D) $W = 900$ R

KT0055

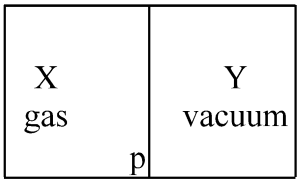
67. An amount Q of heat is added to a monatomic ideal gas in a process in which the gas performs a work $\frac{Q}{2}$ on its surrounding. Find the molar heat capacity for the process.

(A) 1.5 R (B) 2R (C) 3R (D) 4.5 R **KT0315**

68. An ideal gas is taken through a process in which the pressure and the volume are changed according to the equation p = kV. Find the molar heat capacity of the gas for the given process.

(A) $C_v + \frac{R}{2}$ (B) $C_V + R$ (C) $C_v + \frac{R}{3}$ (D) $C_V + 2R$ **KT0316**

69. A closed container is fully insulated from outside. One half of it is filled with an ideal gas X separated by a plate P from the other half Y which contains a vacuum as shown in figure. When P is removed, X moves into Y. Which of the following statements is correct?



(A) No work is done by X (B) X decreases in temperature

(C) X increases in internal energy (D) X doubles in pressure **KT0061**

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70. Water is heated in an open pan where the air pressure is 10^{+5} Pa. The water remains a liquid, which expands by a small amount as it is heated. Determine the ratio of the heat absorbed by the water to the work done by water. (γ for water = 10^{-3}/C, S = 1 cal/gm°C)

(A) 4.2×10^3 (B) 4.2×10^5 (C) 4.2×10^2 (D) 4.2×10^4

KT0062

71. Calculate the increase in internal energy of 1 kg of water at 100 °C when it is converted into steam at the same temperature and at 1 atm (100 kPa). The density of water and steam are 1000 kg m^{-3} and 0.6 kg m^{-3} respectively. The latent heat of vaporization of water = 2.25×10^6 J kg^{-1}.

(A) 1.8×10^6 J (B) 2.08×10^6 J (C) 2.25×10^6 J (D) 3×10^6 J **KT0317**

Thermodynamic cycle

72. A student records ΔQ, ΔU & ΔW for a thermodynamic cycle A → B → C → A. Certain entries are missing. Find correct entry in following options.

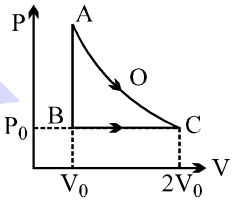
	AB	BC	CA
ΔW	40J		30J
ΔU		50J	
ΔQ	150J	10J	

(A) $W_{BC} = -70$ J (B) $\Delta Q_{CA} = 130$ J

(C) $\Delta U_{AB} = 190$ J (D) $\Delta U_{CA} = -160$ J

KT0056

73. An ideal gas is taken from point A to point C on P-V diagram through two process AOC and ABC as shown in the figure. Process AOC is isothermal



(A) Process AOC requires more heat than process ABC.

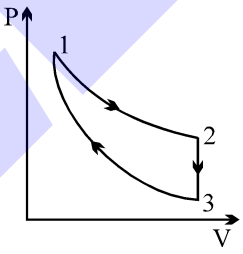
(B) Process ABC requires more heat than process AOC.

(C) Both process AOC & ABC require same amount of heat.

(D) Data is insufficient for comparison of heat requirement for the two processes.

KT0058

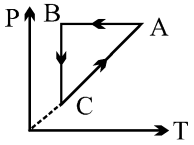
74. Three processes form a thermodynamic cycle as shown on P-V diagram for an ideal gas. Process 1 → 2 takes place at constant temperature (300K). Process 2 → 3 takes place at constant volume. During this process 40J of heat leaves the system. Process 3 → 1 is adiabatic and temperature T_3 is 275K. Work done by the gas during the process 3 →1 is :-

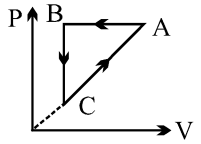


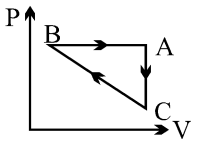
(A) – 40J (B) – 20J (C) +40J (D) +20J

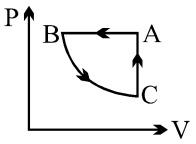
KT0060

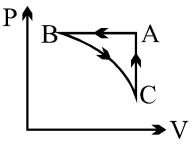
75. A cyclic process ABCA is shown in PT diagram. When presented on PV, it would



(A) 

(B) 

(C) 

(D) 

KT0065

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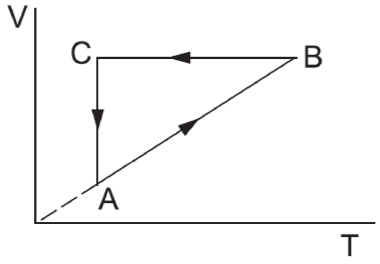
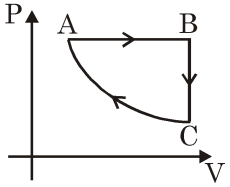
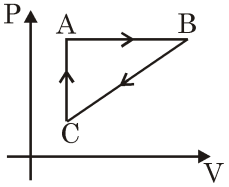
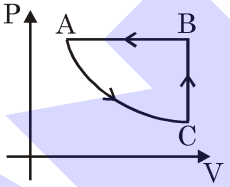
76. A cyclic process ABCA shown in the V–T diagram (figure 24–W3) is performed with a constant mass of an ideal gas. Show the same process on p–V diagram. In the figrue, CA is parallel to the V–axis and BC is paralled to the T-axis.

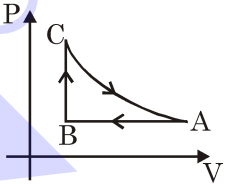


The correct P-V diagram is :- **KT0318**

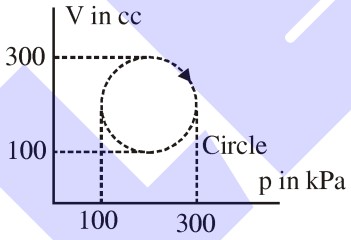
(A) 

(B) 

(C) 

(D) 

77. Calculate the heat lost from the system in going through the cyclic process shown in figure.



(A) 15.7 J (B) 31.4 J (C) 37.68 J (D) 21.98 J **KT0319**

Second Law of Thermodynamics

78. "Heat cannot be itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of- **[AIEEE - 2003]**

(A) second law of thermodynamics (B) conservation of momentum

(C) conservation of mass (D) first law of thermodynamics **KT0068**

79. Which of the following process is reversible :

(A) A block moving on a rough surface comes to rest after same time.

(B) Two gases are mixed in a container

(C) A block kept on a smooth surface is connected to a fixed spring and is oscillating horizontally.

(D) The water is heated on a stove. **KT0320**

80. According to the second law of thermodynamics:

(A) Heat energy cannot be completely converted to work

(B) Work cannot be completely converted to heat energy

(C) For all cyclic processes we have $dQ/T < 0$

(D) The reason all heat engine efficiencies are less than 100% is friction, which is unavoidable

(E) All of the above are true **KT0321**

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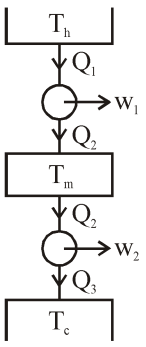
81. A heat engine operates between a cold reservoir at temperature T_2 = 300 K and a hot reservoir at temperature T_1. It takes 200 J of heat from the hot reservoir and delivers 120 J of heat to the cold reservoir in a cycle. What could be the minimum temperature of the hot reservoir ?

(A) 400 K (B) 450 K (C) 500 K (D) 55 K **KT0322**

82. Suppose that two heat engines are connected in series, such that the heat exhaust of the first engine is used as the heat input of the second engine as shown in figure. The efficiencies of the engines are η_1 and η_2, respectively. The net efficiency of the combination is given by



(A) $\eta_{net} = \eta_2 + (1 - \eta_1)\eta_2$ (B) $\eta_{net} = \dfrac{\eta_1}{(1-\eta_1)\eta_2}$

(C) $\eta_{net} = \eta_1 + (1 - \eta_1)\eta_2$ (D) $\eta_{net} = \dfrac{1-\eta_1}{(1-\eta_2)\eta_2}$

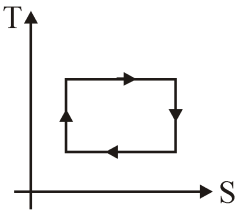
KT0066

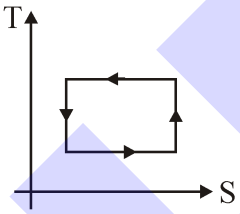
83. A carnot engine takes 3×10^6 cal of heat from a reservoir at 627 °C and gives it to a sink at 27 °C. The work done by the engine is- **[AIEEE - 2003]**

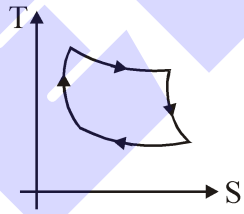
(A) 4.2×10^6 J (B) 8.4×10^6 J (C) 16.8×10^6 J (D) zero

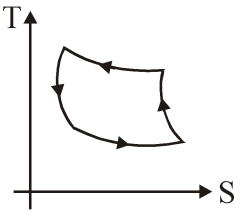
KT0069

84. The carnot cycle for a heat engine on TS diagram is

(A) 

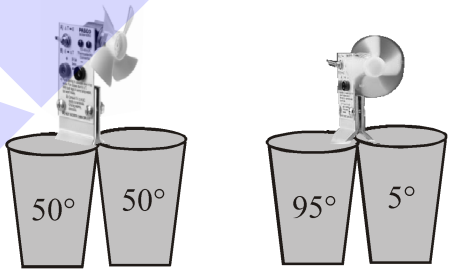
(B) 

(C) 

(D) 

KT0323

85. Here we have a fan running on a thermoelectric converter. It works only when the two electrodes are at different temperature. This can be directly explained by which statement of second law of thermodynamics ?



(A) Carnot's theorem (B) Kelvin plank's statement

(C) Clausius statement (D) $\Delta S_{\text{isolated system}} \geq 0$ **KT0324**

86. The energy entering an electric heater by electrical transmission can be converted to internal energy with an efficiency of 100%. By what factor does the cost of heating your home change when you replace your electric heating system with an electric heat pump that has a COP of 4.00 ? Assume the motor running the heat pump is 100 % efficient.

(A) 4.00 (B) 2.00 (C) 0.500 (D) 0.250 **KT0325**

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87. A heat engine operates with an efficiency of 0.5. What can't the temperatures of the high-temperature and low-temperature reservoirs be?

(A) T_H = 600 K and T_L = 100 K (B) T_H = 600 K and T_L = 200 K

(C) T_H = 500 K and T_L = 200 K (D) T_H = 500 K and T_L = 300 K **KT0326**

88. A refrigerator freezes 5 kg of water at 0°C into ice at 0°C in a time interval of 20 minutes. Assume that room temperature is 20°C. Calculate the minimum power needed to accomplish it ?

(A) 24.4 Watt (B) 0.1025 Watt (C) 0.0244 Watt (D) 102.5 Watt **KT0327**

89. A carnot engine whose source temperature is at 400 K takes 100 kcal of heat at this temperature in each cycle and gives 70 kcal to the sink. Calculate (i) the temperature of the sink and (ii) the efficiency of the engine. **KT0328**

(A) 280 K, 30 % (B) 270 K, 30% (C) 280 K, 10 % (D) None of these

90. An ideal refrigerator operates according to the reverse Carnot cycle and transmits heat from a cold source with water at a temperature of 0°C to a boiler with water at a temperature of 100 °C. What amount of water must be frozen in the cooler to convert 1 kg of water into vapour in the boiler?

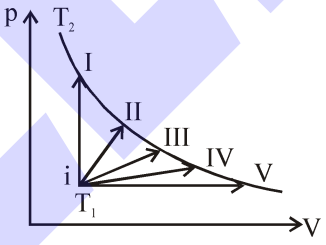
(A) 4.94 kg (B) 3.24 kg (C) 5.63 kg (D) 2.12 kg **KT0329**

91. Oxygen (10 grams) is heated from t_1 = 50°C to t_2 = 150°C. Find the change in entropy if the oxygen is heated: (1) isochorically, (2) isobarically.

(A) ΔS_V = 1.76 J/K; ΔS_P = 2.46 J/K (B) $\Delta S_V = \Delta S_P$ = 1.76 J/K

(C) $\Delta S_V = \Delta S_P$ = 2.46 J/K (D) ΔS_V = 2.46 J/K; ΔS_P = 1.76 J/K **KT0330**

92. An ideal gas is to taken reversibly from state i, at temperature T_1, to any of the other states labeled, I, II, III, IV and V on the p-V diagram below. All are at the same temperature T_2. Rank the five processes according to the change in entropy of the gas, least to greatest :



(A) I, II, III, IV, V (B) V, IV, III, II, I

(C) I, then II, III, IV, and V tied (D) I, II, III and IV tied, then V **KT0331**

93. An inventor claims to have a heat engine that has an efficiency of 40% when it operates between a high temperature reservoir of 150°C and a low temperature reservoir of 30°C. This engine:

(A) Must violate the zeroth law of thermodynamics

(B) Must violate the first law of thermodynamics

(C) Must violate the second law of thermodynamics

(D) Must violate the third law of thermodynamics

(E) Does not necessarily violate any of the laws of thermodynamics **KT0332**

94. One of the two thermally insulated vessels inter connected by a tube with a valve contains 2.2 moles of an ideal gas. The other vessel is evacuated. The valve having been opened, the gas increased its volume n = 3.0 times. Find the approximate entropy increment of the gas.

(A) 10 J/K (B) 12.5 J/K (C) 15 J/K (D) 20 J/K **KT0333**

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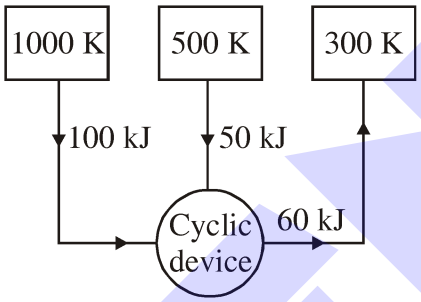
95. A thermally insulated vessel is partitioned into two parts so that the volume of one part is n = 2.0 times greater than that of the other. The smaller part contains $v_1 = 0.30$ mole of nitrogen, and the greater one $v_2 = 0.70$ mole of oxygen. The temperature of the gases is the same. A hole is punctured in the partition and the gases are mixed. Find the corresponding increment of the system's entropy, assuming the gases to be ideal.

(A) 3.4 J/K (B) 5.1 J/K (C) 6.8 J/K (D) 8.5 J/K **KT0334**

96. A cyclic device operates between three thermal reservoirs, as shown in the figure. Heat is transferred to/from the cyclic device. It is assumed that the heat transfer between each thermal reservoir and the cyclic device takes place across negligible temperature difference. Interactions between the cyclic device and the respective thermal reservoirs that are shown in the figure are all in the form of heat transfer**.** The cyclic device can be :-



(A) reversible heat engine.

(B) A reversible heat pump or a refrigerator.

(C) an irreversible heat engine.

(D) an irreversible heat pump or an irreversible refrigerator. **KT0335**

97. Two reversible engines E_1 and E_2 reject heat to a common reservoir at a temperature T. The engine E_1 receives heat from a reservoir at temperature T_1 and delivers work W_1. The Engine E_2 received same amount of heat from a reservoir at temperature T_2 and delivers work W_2. If $T_2 < T_1$, then :-

(A) $W_1 = W_2 = 0$ (B) $W_1 > W_2$ (C) $W_1 < W_2$ (D) $W_1 = W_2$ **KT0336**

98. Which one of the following pairs of equations describes an irreversible heat engine?

(A) $\oint \delta Q > 0$ and $\oint \frac{\delta Q}{T} < 0$ (B) $\oint \delta Q < 0$ and $\oint \frac{\delta Q}{T} < 0$

(C) $\oint \delta Q > 0$ and $\oint \frac{\delta Q}{T} > 0$ (D) $\oint \delta Q < 0$ and $\oint \frac{\delta Q}{T} > 0$ **KT0337**

99. A domestic refrigerator set at 2°C handles on the average a thermal load ot 8000 kJ/hour. The ambient temperature is 30°C. The COP ot the refrigerator is equal to 0.15 times that of a Carnot refrigerator. The electricity consumption (in kW) is approximately

(A) 1.47 (B) 1.51 (C) 3.28 (D) 2.86 **KT0338**

100. Coefficient of performance of a Reversed carnot cycle refrigerator working between higher temperature T_2 and lower temperature T_1.

(A) Will increase with increase in T_1 keeping T_2 fixed

(B) Will decrease with increase in T_1 keeping T_2 fixed

(C) Will first increase with increase in T_1 and then decrease with increase in T_1 keeping T_2 fixed

(D) None of the above **KT0339**

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101. A refrigerating machine working on reversed carnot cycle takes out 2 kW of heat from the system while working between temperature limits of 300 K and 200 K. COP and power consumed by the cycle will be respectively
(A) l and 1 kW (B) l and 2kW (C) 2 and 1 kW (D) 2 and 2 kW **KT0340**

102. Process 'A' is irreversible and adiabatic. Process 'B' is reversible and adiabatic. The entropy change in process 'A' and process 'B' respectively are
(A) zero and positive (B) zero and negative (C) Negative and zero (D) Positive and zero **KT0341**

103. The approximate entropy change, when 10 kg of an ideal gas having specific heat at constant volume $c_v = 5R/2$ (R = 287 J/kg-K) is taken from an initial state of 100 kPa and 300 K to the final state of 200 kPa and 500 K is :-
(A) 9.1 kJ/K (B) 3.14 kJ/K (C) 9.10 kJ/K (D) 0.314 kJ/K **KT0342**

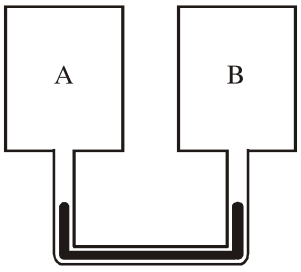
104. Consider an isentropic process undergone by an incompressible liquid. The change in temp. experiment by the liquid is $\Delta T = T_2 - T_1$. Which one of the fallowing is correct?
(A) $\Delta T = 0$
(B) $\Delta T > 0$
(C) $\Delta T < 0$
(D) ΔT depends on liquid under consideration **KT0343**

105. A heat engine operates between two thermal reservoirs one at 100°C and another at 20° C. 373 J of energy is removed from the hot reservoir. The change in entropy of the hot reservoir in J/k is
(A) – 1.27 (B) 1.0 (C) –1.0 (D) 1.27 **KT0344**

MULTIPLE CORRECT TYPE QUESTIONS

106. On the P–T graph of an ideal gas,
(A) adiabatic process will be a straight line
(B) isochoric process will be a straight line passing through the origin
(C) adiabatic curve will have a positive slope
(D) the slope of adiabatic curve will decrease with increase in T **KT0071**

107. Figure shows two rigid vessels A and B, each of volume 200 cm^3 containing an ideal gas (C_v = 12.5 J/mol–K). The vessels are connected to a manometer tube containing mercury. The pressure in both the vessels is 75 cm of mercury and the temperature is 300 K. 5.0 J of heat is supplied to the gas in the vessel A and 10 J to the gas in the vessel B. (Assuming no appreciable transfer of heat from A to B. Gas constant R = 8.3 J/mol–K.)



(A) The number of moles of the gas in each vessel 0.008
(B) The number of moles of the gas in each vessel 0.004
(C) Difference in mercury level on two sides of manometer is 12.5 cm.
(D) Difference in mercury level on two sides of manometer is 10.0 cm. **KT0345**

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108. Figure shows an adiabatic cylindrical tube of volume V_0 divided in two parts by a frictionless adiabatic separator. Initially, the separator is kept in the middle, an ideal gas at pressure p_1 and temperature T_1 is injected into the left part and another ideal gas at pressure p_2 and temperature T_2 is injected into the right part. $\frac{C_p}{C_v} = \gamma$ is the same for both the gases. The separator is slid slowly and is released at a position where it can stay in equilibrium. Find

(A) The volumes of the two parts are $\frac{p_1^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ and $\frac{p_2^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ respectively.

(B) The volumes of the two parts are $\frac{p_2^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ and $\frac{p_1^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ respectively.

(C) The heat given to the gas in the left part is zero.

(D) The final common pressure of the gases is $\left(\frac{p_1^{1/\gamma}+p_2^{1/\gamma}}{2}\right)^{\gamma}$ **KT0346**

109. Two gases have the same initial pressure, volume and temperature. They expand to the same final volume, one adiabatically and the other isothermally

(A) The final temperature is greater for the isothermal process

(B) The final pressure is greater for the isothermal process

(C) The work done by the gas is greater for the isothermal process

(D) All the above options are incorrect. **KT0070**

110. A mixture of ideal gases 7 kg of nitrogen and 11 kg of CO_2. Then

(A) equivalent molecular weight of the mixture is 36.

(B) equivalent molecular weight of the mixture is 18.

(C) γ for the mixture is 5/2

(D) γ for the mixture is 47/35.

(Take γ for nitrogen and CO_2 as 1.4 and 1.3 respectively) **KT0347**

111. The ratio of the molar heat capacities of an ideal gas is $\frac{C_p}{C_v} = \frac{7}{6}$. Calculate the change in internal energy of 1.0 mole of the gas when its temperature is raised by 50 K. (R = 8.3 J/mol-K)

(i) keeping the pressure constant .

(ii) keeping the volume constant and

(iii) adiabatically

(A) 2400 J, 2490 J & 2500 J

(B) 2490 J in all cases

(C) Always unequal for different processes.

(D) ΔU is independent of process for given change in temperature. **KT0348**

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112. Three identical adiabatic containers A, B and C contain helium, neon and oxygen respectively at equal pressure. The gases are pushed to half their original volumes.
(A) The final temperature in the three containers will be the same.
(B) The final pressures in the three containers will be the same.
(C) The pressure of helium and neon will be the same but that of oxygen will be different.
(D) The temperature of helium and neon will be the same but that of oxygen will be different
KT0349

113. An ideal gas expands from 100 cm^3 to 200 cm^3 at a constant pressure of 2.0×10^5 Pa when 50 J of heat is supplied to it.
(A) The change in internal energy of the gas 30 J.
(B) The number of moles in the gas if the initial temperature is 300 K is 0.008.
(C) The molar heat capacity C_p at constant pressure is 20.8 JK^{-1} mol^{-1}
(D) The molar heat capacity C_v at constant volume is 12.5 JK^{-1} mol^{-1} **KT0350**

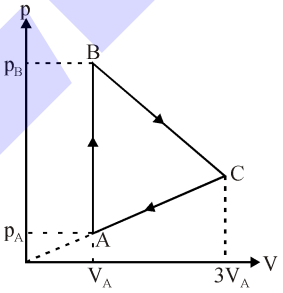
114. During the melting of a slab of ice at 273 K at atmospheric pressure
(A) Positive work is done by the ice-water system on the atmosphere
(B) Positive work is done on the ice-water system by the atmosphere
(C) The internal energy of the ice-water increases
(D) The internal energy of ice-water system decreases **KT0072**

115. The volume of an ideal gas ($\gamma = 1.5$) is changed adiabatically from 4.00 litres to 3.00 litres.
(A) The ratio of final pressure to the initial pressure is 1.54.
(B) The ratio of final pressure to the initial pressure is 1.1.
(C) The ratio of final temperature to the initial temperature is 1.15.
(D) The ratio of final temperature to the initial temperature is 1.5. **KT0351**

COMPREHENSION TYPE QUESTIONS

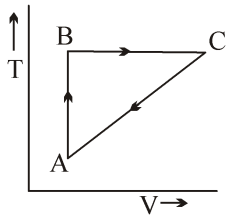
Paragraph for Question nos. 116 and 117

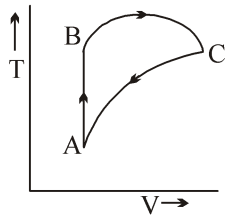
A sample of ideal gas is taken through the cyclic process shown in the figure. The temperature of the gas at state A is $T_A = 200$ K. At states B and C the temperature of the gas is the same.

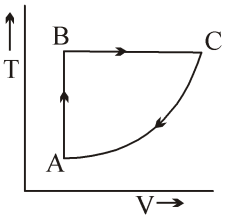


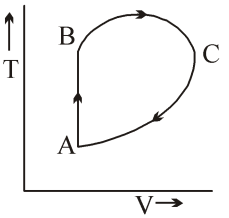
116. Net work done by the gas in the process is
(A) $2P_AV_A$ (B) $4P_AV_A$ (C) $6P_AV_A$ (D) $8P_AV_A$
KT0073

117. Which of the following graphs best represent the cyclic process in T-V diagram.

(A) 

(B) 

(C) 

(D) 

KT0073

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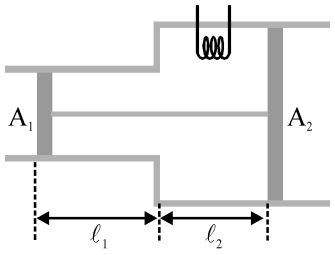
Paragraph for Question Nos. 118 to 120

The walls of the two connecting cylinders shown in the figure are adiabatic (thermally insulating). The cross-sectional areas of the parts are A_1 and A_2. There is a well-fitting but freely moving, thermally insulating piston in each cylinder, at a distance $\ell_1 = \ell_2 = \ell$ from the point where the cross-sectional area changes. The pistons are connected to each other by a thin and rigid rod. The enclosed volume contains air. The temperature and air pressure are T_0 and p_0 both inside and outside. The heater filament inside is operated for time t at a power of P.



118. Mark the **CORRECT** statement :-

(A) Process performed is adiabatic
(B) Piston shifts towards left
(C) Process performed is polytropic
(D) Piston shifts towards right

KT0074

119. What is change in temperature (Q is heat energy supplied, v_0 is initial volume) :-

(A) $\Delta T = \frac{2}{5}\frac{QT_0}{p_0V_0}$ (B) $\Delta T = \frac{2}{7}\frac{QT_0}{p_0V_0}$ (C) $\Delta T = \frac{5}{7}\frac{QT_0}{p_0V_0}$ (D) $\Delta T = \frac{QT_0}{p_0V_0}$

KT0074

120. What is displacement (x) of piston till equilibrium is reached :-

(A) $x = \frac{2Pt}{(A_2 - A_1)7p_0}$ (B) $x = \frac{2Pt}{(A_2 - A_1)5p_0}$

(C) $x = \frac{2Pt}{(A_2 - A_1)3p_0}$ (D) $x = \frac{2Pt}{(A_2 - A_1)9p_0}$

KT0074

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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

KTG

1. An open and wide glass tube is immersed vertically in mercury in such a way that length 0.05 m extends above mercury level. The open end of the tube is closed and the tube is raised further by 0.43m. The length of air column above mercury level in the tube will be : Take P_{atm} = 76 cm of mercury

(A) 0.215 m (B) 0.2 m (C) 0.1 m (D) 0.4 m

KT0076

2. A barometer tube, containing mercury, is lowered in a vessel containing mercury until only 50 cm of the tube is above the level of mercury in the vessel. If the atmospheric pressure is 75 cm of mercury, what is the pressure at the top of the tube ?

(A) 33.3 kPa (B) 66.7 kPa (C) 3.33 MPa (D) 6.67 MPa

KT0077

3. At temperature T, N molecules of gas A each having mass m and at the same temperature 2N molecules of gas B each having mass 2m are filled in a container. The mean square velocity of molecules of gas B is v^2 and mean square of x component of velocity of molecules of gas A is w^2. The ratio of w^2/v^2 is :

(A) 1 (B) 2 (C) 1/3 (D) 2/3

KT0078

4. A gas is enclosed in a vessel at a constant temperature at a pressure of 5 atmosphere and volume 4 litre. Due to a leakage in the vessel, after some time, the pressure is reduced to 4 atmosphere. As a result, the

(A) volume of the gas decreased by 20%.

(B) average K.E. of gas molecule decreases by 20%.

(C) 20% of the gas escaped due to the leakage.

(D) 25% of the gas escaped due to the leakage. **KT0087**

5. Two closed vessels of equal volume contain air at 105 kPa, 300 K and are connected through a narrow tube. If one of the vessels is now maintained at 300 K and the other at 400 K, what will be the pressure in the vessels ?

(A) 100 kPa (B) 120 kPa (C) 150 kPa (D) 160 kPa **KT0352**

6. A horizontal tube of length ℓ closed at both ends contains an ideal gas of molecular weight M. The tube is rotated at a constant angular velocity ω about a vertical axis passing through an end. (Assume the temperature to be uniform and constant. p_2 and p_1 denote the pressures at the free end and the fixed end respectively.)

(A) $p_2 = p_1 e^{\frac{M\omega^2\ell^2}{2RT}}$ (B) $p_2 = p_1 e^{\frac{M\omega^2\ell^2}{RT}}$ (C) $p_2 = p_1 e^{\frac{M\omega^2\ell^2}{3RT}}$ (D) $p_2 = p_1 e^{\frac{M\omega^2\ell^2}{4RT}}$

KT0353

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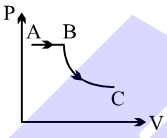
7. A vertical cylinder of height 100 cm contains air at a constant temperature. The top is closed by a frictionless light piston. The atmospheric pressure is equal to 75 cm of mercury. Mercury is slowly poured over the piston. Find the maximum height of the mercury column that can be put on the piston.
(A) 20 cm (B) 25 cm (C) 27.5 cm (D) 22.5 cm **KT0354**

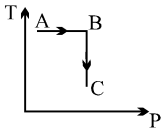
8. A cylinder made of perfectly non conducting material closed at both ends is divided into two equal parts by a heat proof piston. Both parts of the cylinder contain the same masses of a gas at a temperature $t_0 = 27°$ C and pressure $P_0 = 1$ atm. Now if the gas in one of the parts is slowly heated to t = 57°C while the temperature of first part is maintained at t_0 the distance moved by the piston from the middle of the cylinder will be (length of the cylinder = 84 cm)
(A) 3 cm (B) 5 cm (C) 2 cm (D) 1 cm **KT0081**

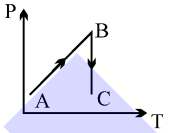
First law of thermodynamics

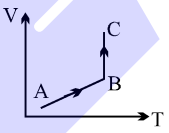
9. 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :
(A) 32 kJ (B) 5 kJ (C) – 4 kJ (D) 13 kJ **KT0082**

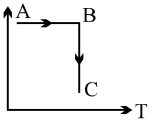
10. A process is shown in the diagram. Which of the following curves may represent the same process?



(A) 

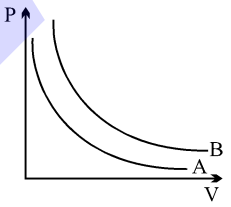
(B) 

(C) 

(D) 

KT0083

11. Figure shows the pressure P versus volume V graphs for two different gas sample at a given temperature. M_A and M_B are masses of two samples, n_A and n_B are numbers of moles. Which of the following **must be incorrect.**



(A) $M_A > M_B$ (B) $M_A < M_B$ (C) $n_A > n_B$ (D) $n_A < n_B$

KT0085

12. A vertical cylinder with heat-conducting walls is closed at the bottom and is fitted with a smooth light piston. It contains one mole of an ideal gas. The temperature of the gas is always equal to the surrounding's temperature, T_0. The piston is moved up slowly to increase the volume of the gas to η times. Which of the following is incorrect?
(A) Work done by the gas is $RT_0 \ln \eta$.
(B) Work done against the atmosphere is $RT_0(\eta - 1)$.
(C) There is no change in the internal energy of the gas.
(D) The final pressure of the gas is $\frac{1}{(\eta - 1)}$ times its initial pressure. **KT0086**

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13. The value of Cp - Cv is 1.00 R for a gas sample in state A and is 1.08 R in state B. Let p_A, p_B denote the pressure and T_A and T_B donote the temperature of the states A and B respectively. Most likely

(A) $p_A < p_B$ and $T_A > T_B$ (B) $p_A > p_B$ and $T_A < T_B$

(C) $p_A = p_B$ and $T_A < T_B$ (D) $p_A > p_B$ and $T_A = T_B$ **KT0355**

14. In Joly's differential steam calorimeter, 3 g of an ideal gas is contained in a rigid closed sphere at 20°C. The sphere is heated by steam at 100°C and it is found that an extra 0.095 g of steam has condensed into water as the temperature of the gas becomes constant. Calculate the specific heat capacity of the gas in J/g-K. The latent heat of vaporization of water = 540 cal/g.

(A) 0.60 J/gK (B) 0.80 J/gK (C) 0.90 J/gK (D) 1.00 J/gK **KT0356**

15. An one mole of monatomic ideal gas is contained in a rigid container of volume V with walls of total inner surface area A, thickness x and thermal conductivity K. The gas is at an initial temperature T_0 and pressure p_0 . Find the pressure of the gas as a function of time if the temperature of the surrounding air is T_s. All temperatures are in absolute scale.

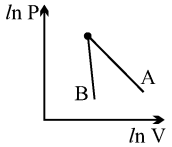
(A) $\frac{p_0}{T_0}\left[T_S-(T_S-T_0)e^{-\frac{3KA}{2xR}t}\right]$ (B) $\frac{p_0}{T_0}\left[T_S-(T_S-T_0)e^{-\frac{KA}{3xR}t}\right]$

(C) $\frac{p_0}{T_0}\left[T_S-(T_S-T_0)e^{-\frac{2KA}{xR}t}\right]$ (D) $\frac{p_0}{T_0}\left[T_S-(T_S-T_0)e^{-\frac{2KA}{3xR}t}\right]$ **KT0357**

16. Figure shows a cylindrical tube with perfect conducting walls and fitted with perfect conducting separator. The separator can be slid in the tube by an external mechanism. An ideal gas is injected in the two sides at equal pressure and equal temperatures. The separator remains in equilibrium at the middle. It is now slid to a position where it divides the tube in the ratio of 1:3. Find the ration of the presure in the two parts of the vessel.

(A) 1 : 2 (B) 3 : 1 (C) 2 : 3 (D) 4 : 1 **KT0358**

17. The figure, shows the graph of logarithmic reading of pressure and volume for two ideal gases A and B undergoing adiabatic process. From figure it can be concluded that



(A) gas B is diatomic (B) gas A and B both are diatomic

(C) gas A is monoatomic (D) gas B is monoatomic & gas A is diatomic

KT0079

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18. Air ($\gamma = 1.4$) is pumped at 2 atm pressure in a motor tyre at 20°C. If the tyre suddenly bursts, what would be the temperature of the air coming our of the tyre. Neglect any mixing with the atmospheric air.
(A) 240 K (B) 250 K (C) 280 K (D) 300 K **KT0359**

19. An ideal gas expands in such a way that PV^2 = constant throughout the process.
(A) The graph of the process of T-V diagram is a parabola.
(B) The graph of the process of T-V diagram is a straight line.
(C) Such an expansion is possible only with heating.
(D) Such an expansion is possible only with cooling. **KT0084**

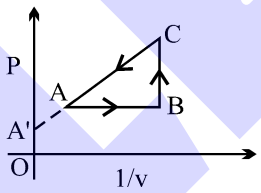
20. When unit mass of water boils to become steam at 100^0C, it absorbs Q amount of heat. The densities of water and steam at 100^0C are ρ_1 and ρ_2 respectively and the atmospheric pressure is p_0. The increase in internal energy of the water is

(A) Q (B) $Q + p_0\left(\frac{1}{\rho_1} - \frac{1}{\rho_2}\right)$

(C) $Q + p_0\left(\frac{1}{\rho_2} - \frac{1}{\rho_1}\right)$ (D) $Q - p_0\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)$ **KT0080**

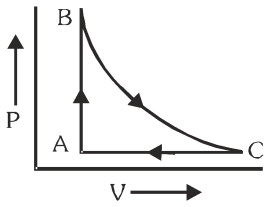
Thermodynamic cycle

21. An enclosed ideal gas is taken through a cycle as shown in the figure. Then



(A) Along AB, temperature decreases while along BC temperature increases
(B) Along AB, temperature increases while along BC the temperature decreases.
(C) Along CA work is done by the gas and the internal energy remains constant.
(D) Along CA work is done on the gas and internal energy of the gas increases. **KT0088**

22. One mole of monoatomic ideal gas undergoes a cyclic process ABCA as shown in figure. Process BC is adiabatic. The temperatures at A, B and C are 300, 600 and 450K respectively. Choose the correct statement(s).



(A) In process CA change in internal energy is 225R.
(B) In process AB change in internal energy is –150R.
(C) In process BC change in internal energy is –225R.
(D) Change in internal energy during the whole cyclic process is +150R. **KT0089**

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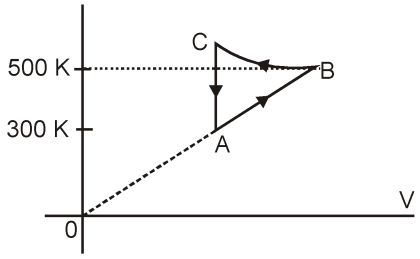
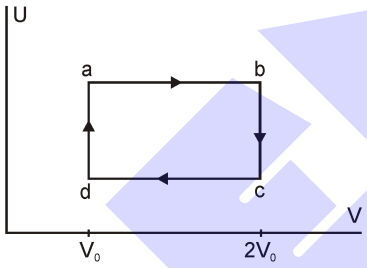
23. Consider the cyclic process ABCA, shown in figure, performed on a sample of 2.0 mole of an ideal gas. A total of 1200 J of heat is withdrawn from the sample in the process. Find the work done by the gas during the part BC.



(A) –4480 J (B) –4520 J (C) –4540 J (D) –4460 J **KT0360**

24. Figure shown the variation in the internal energy U with the volume V of 2.0 mole of an ideal gas in a cyclic process abcda. The temperature of the gas at b and c are 500 K and 300 K respectively. Calculate the heat absorbed by the gas during the process.



(A) 2300 J (B) 2200 J (C) 2400 J (D) 2500 J **KT0361**

Second law of thermodynamics

25. The change in entropy between the two adiabatic lines of a Carnot cycle is 1 kcal/deg. The temperature difference between the two isothermal lines is equal to 100°C. What amount of heat will be converted into work during this cycle?

(A) 420 kJ (B) 840 kJ (C) 210 kJ (D) 105 kJ **KT0362**

26. A carnot heat engine runs between a cold reservoir at temperature T_C and a hot reservoir at temperature T_H. You want to increase its efficiency. Of the following, which change results in the greatest increase in efficiency? The value of ΔT is the same for all changes.

(A) Raise the temperature of the hot reservoir by ΔT
(B) Raise the temperature of the cold reservoir by ΔT
(C) Lower the temperature of the hot reservoir by ΔT
(D) Lower the temperature of the cold reservoir by ΔT
(E) Lower the temperature of the hot reservoir by $1/2\Delta T$ and raise the temperature of the cold reservoir by $1/2\Delta T$. **KT0363**

27. The specific heat capacity of a metal at low temperautre (T) is given as $C_p(kJk^{-1}kg^{-1}) = 32\left(\frac{T}{400}\right)^3$

A 100 gram vessel of this metal is to be cooled from 20°K to 4°K by a special refrigerator operating at room temperature (27°C). The amount of work required to cool the vessel is: **[AIEEE-2011]**

(A) equal to 0.002 kJ (B) greater than 0.148 kJ
(C) between 0.148 kJ and 0.028 kJ (D) less than 0.028 kJ

KT0105

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28. An ideal gas was expanded from the initial state to the volume V without any heat exchange with the surrounding bodies. Will the final gas pressure be the same in the case of (i) a fast and in the case of (ii) a very slow expansion process ?

(A) Lower in fast expansion (B) Higher is fast expansion

(C) Equal in both (D) Cannot be determined **KT0364**

29. A heat engine operates b/w three reservoir, R_1 at 550 K, R_2 at 450 K and R_3 at 350 K. For every cycle, the engine accepts 100 kJ from R_1 and rejects 60 kJ in R_2 and 30 kJ into R_3. The engine efficiency is:

(A) 0.10 (B) 0.20 (C) 0.30 (D) can not be defined

KT0365

30. The working temperatures in the evaporator and condenser coils of a refrigerator are –30°C and 32°C respectively. If the actual refrigerator has COP of 0.75 of the maximum, the required power input for refrigerating effect of 5kW is, nearly

(A) 1.7kW (B) 2.94 kW (C) 3.92 kW (D) 4.0 kW **KT0366**

31. A solar energy based heat engine which reveives 80 kJ of heat at 100°C and rejects 70 kJ of heat to the ambient at 30°C is to be designed. The thermal efficiency of the heat engine is

(A) 70% (B) 18.8% (C) 12.5% (D) Indetermine **KT0367**

32. A metallic object cools from temperature T_1 to temperature T_2. The specific heat of object varies as $C = a + bT^2$ (where T is in kelvin). The change in specific entropy of the object during the cooling process is:

(A) $a\ln\left(\frac{T_2}{T_1}\right)+0.5b\left(T_2^2-T_1^2\right)$ (B) $a\ln\left(\frac{T_2}{T_1}\right)$

(C) $a\ln\left(T_1+T_2\right)+0.5b\left(T_2^2-T_1^2\right)$ (D) $a\ln\left[a+b\left(\frac{T_1+T_2}{2}\right)\right]$ **KT0368**

MULTIPLE CORRECT TYPE QUESTIONS

33. Let v_{av}, v_{rms} and v_p respectively denote mean speed, root mean square speed and the most probable speed of the molecule in an ideal monoatomic gas at absolute temperature T. The mass of a molecule is m then :

(A) no molecule can have speed greater than $\sqrt{2}\,v_{rms}$

(B) no molecule can have speed less than $v_p/\sqrt{2}$

(C) $v_p < v_{av} < v_{rms}$

(D) the average kinetic energy of a molecule is 3/4 mv_p^2

KT0090

34. A gas expands such that its initial and final temperature are equal. Also, the process followed by the gas traces a straight line on the P-V diagram :

(A) The temperature of the gas remains constant throughout.

(B) The temperature of the gas first increases and then decreases

(C) The temperature of the gas first decreases and then increases

(D) The straight line has a negative slope. **KT0091**

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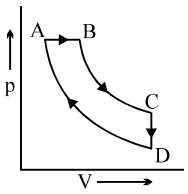
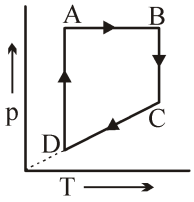
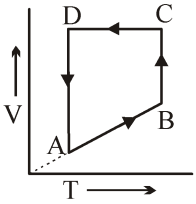
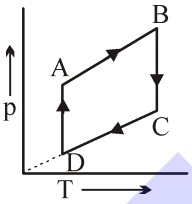
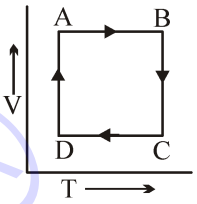
35. A cyclic process ABCD is shown in the p-V diagram. Which of the following curves represents the same process if BC & DA are isothermal processes



(A) 

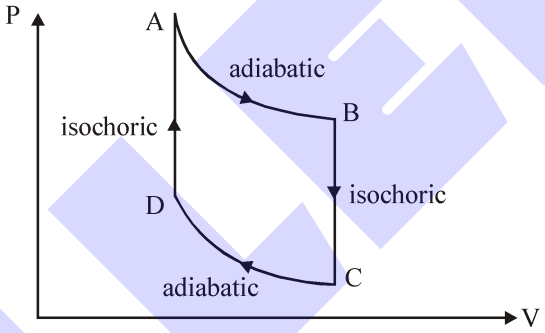
(B) 

(C) 

(D) 

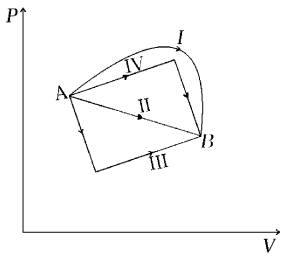
KT0092

36. Two moles of helium gas is taken through the cycle ABCDA as shown in the figure. If $T_A = 1000$ K, $2P_A = 3P_B = 6P_C$.



(A) work done by the gas in the process A to B is 3741 J.
(B) heat lost by the gas in the process B to C is 10600 J.
(C) temperature T_D is 2000 K.
(D) none of these **KT0093**

37. Figure shows the *P-V* diagram of an ideal gas undergoing a change of state from A to B. Four different parts I, II, III and IV as shown in the figure may lead to the same change of state.



(A) Change in internal energy is same in IV and III cases, but not in I and II.
(B) Change in internal energy is same in all the four cases.
(C) Work done is maximum in case I
(D) Work done is minimum in case II. **KT0094**

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38. Two bodies A & B have equal mass m and same specific heat capacity S. They are at temperature T and 4T respectively.

(A) If they are brought in contact, change in entropy of universe will be $ms\ell n\left(\frac{25}{16}\right)$.

(B) If they are used as sink and source for a reversible heat engine, the maximum work obtainable is mCT.

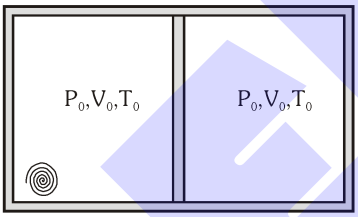
(C) For option B, final temperature is $\frac{5T}{2}$.

(D) For option B, final temperature is 2T. **KT0369**

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 39 to 41

One mole of a monoatomic ideal gas occupies two chambers of a cylinder partitioned by means of a movable piston. The walls of the cylinder as well as the piston are thermal insulators. Initially equal amounts of gas fill both the chambers at (P_0, V_0, T_0). A coil is burnt in the left chamber which absorbs heat and expands, pushing the piston to the right. The gas on the right chamber is compressed until to pressure becomes 32 P_0.



39. The final volume of left chamber is :

(A) $\frac{V_0}{8}$ (B) $\frac{15}{8}V_0$ (C) $\frac{7}{8}V_0$ (D) $\frac{9}{8}V_0$

KT0095

40. The work done on the gas in the right chamber is :-

(A) $\frac{9}{2}P_0V_0$ (B) $-\frac{9}{2}P_0V_0$ (C) $\frac{13}{2}P_0V_0$ (D) $\frac{17}{2}P_0V_0$

KT0095

41. The change in internal energy of the gas in the left chamber is :-

(A) $\frac{186}{4}RT_0$ (B) $\frac{177}{4}RT_0$ (C) $\frac{59}{2}RT_0$ (D) $\frac{131}{4}RT_0$

KT0095

Paragraph for Question no. 42 & 43

An air conditioner installed in a room is operated by carnot cycle in reverse. It acts like a carnot refrigerator. The room is closed from all sides. The walls lose heat to surrounding by conduction. The inside of room is at 27°C when outside is at 37°C. The AC consumes a power of 1000W in this case.

42. If the outside temperature increases to 47°C, what will be the power consumed if the inside temperature is still 27°C.

(A) 2000W (B) 3000W (C) 4000W (D) 8000W **KT0370**

43. If the outside temperature increases to 47°C, but the AC still operates at 1000W, what will be the inside temperature (approx.)

(A) 30°C (B) 32°C (C) 34°C (D) 37°C **KT0370**

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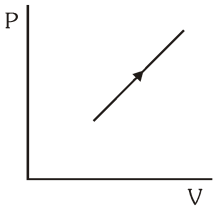
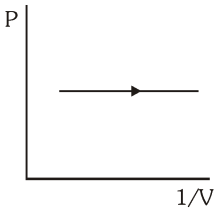
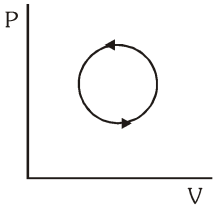
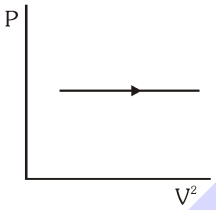
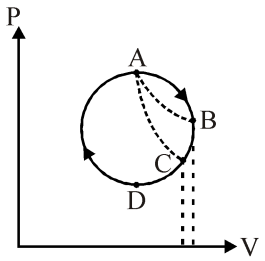
MATRIX MATCH TYPE QUESTION

44.

	Column I (Pressure volume graph)	**Column II (W is work done by gas, Q heat supply to gas)**
(A)		(P) $W > 0$
(B)		(Q) $W < 0$
(C)		(R) $Q > 0$
(D)		(S) $Q < 0$

KT0096

45. For an ideal gas a process PV diagram is a circle. An adiabatic from A passes through C. An isotherm from A passes through B. We take a part of the circular cyclic process. Comment on the sign of the quantity of column-I.



	Column-I		**Column-II**
(A)	Heat given to the gas in going from A to C along circle	(P)	Positive
(B)	Heat given to the gas in going from B to C along circle	(Q)	Negative
(C)	Heat given to the gas in going from C to D along circle	(R)	Zero
		(S)	can't be said

KT0075

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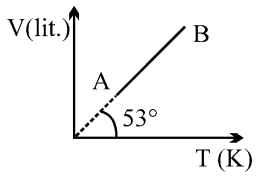
EXERCISE (S)

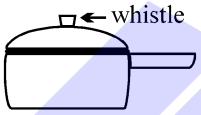
KTG

1. V-T curve for 2 moles of a gas is straight line as shown in the graph here. Find the pressure of gas at A.



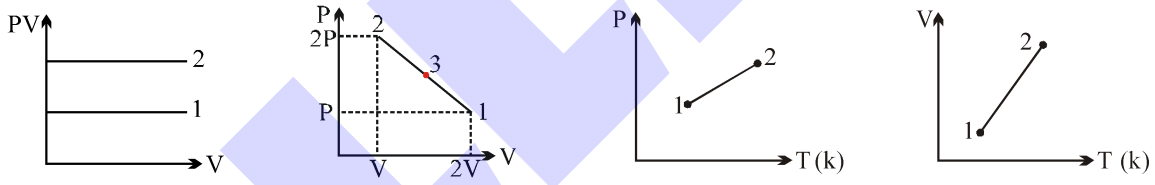
KT0001

2. An empty pressure cooker of volume 10 litres contains air at atmospheric pressure 10^5 Pa and temperature of 27°C. It contains a whistle which has area of 0.1 cm^2 and weight of 100 g. What should be the temperature of air inside so that the whistle is just lifted up?



KT0002

3. Examine the following plots and predict whether in
(i) $P_1 < P_2$ and $T_1 > T_2$, in (ii) $T_1 = T_2 < T_3$,in (iii) $V_1 > V_2$, in (iv) $P_1 > P_2$ or otherwise.



(i) (ii) (iii) (iv)

KT0003

4. A glass tube, sealed at both ends, is 100 cm long. It lies horizontally with the middle 10 cm containing mercury. The two ends of the tube contain air at 27^0C and at a pressure 76 cm of mercury. The air column on one side is maintained at 0^0C and the other side is maintained at 127^0C. Calculate the length of the air column on the cooler side. Neglect the changes in the volume of mercury and of the glass.

KT0371

5. A barometer is faulty. When the true barometer reading are 73 and 75 cm of Hg, the faulty barometer reads 69 cm and 70 cm respectively.
(i) What is the total length of the barometer tube?
(ii) What is the true reading when the faulty barometer reads 69.5 cm ?
(iii) What is the faulty barometer reading when the true barometer reads 74 cm? **KT0017**

6. Modern vacuum pumps permit the pressures down to $p = 4 \times 10^{-15}$ Pa to be reached at room temperatures(300 k). Assuming that the gas exhausted is nitrogen, find the number of its molecules per cm^3 and the mean distance between them at this pressure. Also find then mean free path. [Take $\sigma = 4$Å.] **KT0372**

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7. The temperature of a gas consisting of rigid diatomic molecules is T = 300 K. Calculate the angular root mean square velocity of a rotating molecule if its moment of inertia is equal to $I = 2.1 \times 10^{-39}$ g.cm^2. **KT0373**

8. An ideal gas of molar mass M is contained in a tall vertical cylindrical vessel whose base area is S and height h. The temperature of the gas is T, its pressure on the bottom base is p_0. Assuming the temperature and the free-fall acceleration g to be independent of the height, find the mass of gas in the vessel. **KT0374**

9. A tall cylindrical vessel with gaseous nitrogen is located in a uniform gravitational field in which the free-fall acceleration is equal to g. The temperature of the nitrogen varies along the height h so that its density is the same throughout the volume. Find the temperature gradient dT/dh. **KT0375**

10. Hydrogen gas is contained in a closed vessel at 1 atm (100 kPa) and 300K

(a) Calculate the mean speed of the molecule.

(b) Suppose the molecules strike the wall with this speed making an average angle of 45° with it. How many molecule strike each square meter of the wall per second? **KT0376**

11. Show that the internal energy of the air (treated as an ideal gas) contained in a room remains constant as the temperature changes between day and night. Assume that the atmospheric pressure around remains constant that the air in the room maintains this pressure by communicating with the surrounding through the windows etc. **KT0377**

12. Suppose a gas is heated up to a temperature at which all degrees of freedom (translational, rotational, and vibrational) of its molecules are excited. Find the molar heat capacity of such a gas in the isochoric process, if the gas consists of

(a) diatomic;

(b) linear having 4 atmos in molecule

(c) non linear having 4 atmos in the molecule **KT0378**

13. A mixture of 4 gm helium and 28 gm of nitrogen is enclosed in a vessel of constant volume 300 K. Find the quantity of heat absorbed by the mixture to doubled the root mean velocity of its molecules. (R = Universal gas constant) **KT0006**

14. An insulated container containing monoatomic gas of molar mass m is moving with a velocity v_0. If the container is suddenly stopped, find the change in temperature. **[JEE 2003]** **KT0007**

15. If heat is added at constant volume, 6300 J of heat are required to raise the temperature of an ideal gas by 150 K. If instead, heat is added at constant pressure, 8800 joules are required for the same temperature change. When the temperature of the gas changes by 300 K, determine the change is the internal energy of the gas. **KT0008**

16. Find the molecular mass of a gas if the specific heats of the gas are C_P=0.2 cal/gm°C and C_V = 0.15 cal/gm°C. [Take R = 2 cal/mole°C] **KT0004**

17. In changing the state of a gas adiabatically from an equilibrium state A to another equilibrium state B, an amount of work equal to 22.3 J is done on the system. If the gas is taken from state A to B via a process in which the net heat absorbed by the system is 9.35 cal, how much is the net work done by the system in the latter case ? (Take 1 cal = 4.19 J) **KT0013**

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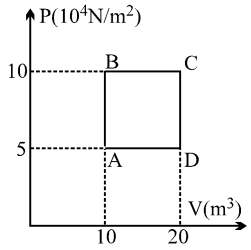
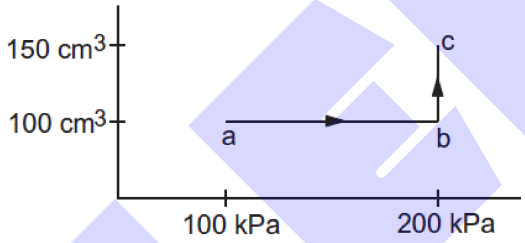
18. A sample of 2 kg of monoatomic Helium (assumed ideal) is taken through the process ABC and another sample of 2 kg of the same gas is taken through the process ADC as in figure. Given, molecular mass of Helium = 4



(i) What is the temperature of Helium in each of the states A, B, C & D ?
(ii) Is there any way of telling afterwards which sample of Helium went through the process ABC and which went through the process ADC? Write Yes or No.
(iii) How much is the heat involved in each of the processes ABC & ADC. **KT0026**

19. An ideal gas ($\gamma = 1.67$) is taken through the process abc shown in figure. The temperature at the point a is 300 K. Calculate



[a] the temperature at b and c,
[b] the work done in the process,
[c] the amount of heat supplied in the path ab and in the path bc and
[d] the change in the internal energy of the gas in the process. **KT0379**

20. A cylinder with a movable piston contains 3 moles of hydrogen at standard temperature and pressure. The walls of the cylinder are made of a heat insulator, and the piston is insulated by having a pile of sand on it. By what factor does the pressure of the gas increase if the gas is compressed to half its original volume ? **KT0010**

21. Two moles of helium gas ($\gamma = 5/3$) are initially at 27°C and occupy a volume of 20 litres. The gas is first expanded at constant pressure until the volume is doubled. Then it undergoes an adiabatic change until the temperature returns to its initial value. (a) Sketch the process in a p–V diagram. (b) What is the final volume and pressure of the gas ? (c) What is the work done by the gas ? **KT0380**

22. A gas is enclosed in a cylindrical can fitted with a piston. The walls of the can and the piston are adiabatic. The initial pressure, volume and temperature of the gas are 100 kPa, 400 cm^3 and 300 K respectively. The ratio of the specific heat capacities of the gas are 100 kPa, 400 cm^3 and 300 K respectively. The ratio of the specific heat capacities of the gas is $\frac{C_p}{C_v} = 1.5$. Find the pressure and the temperature of the gas if it is
[a] suddenly compressed
[b] slowly compressed to 100 cm^3. **KT0381**

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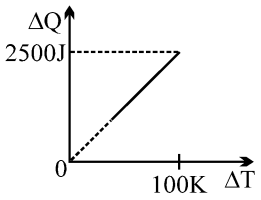
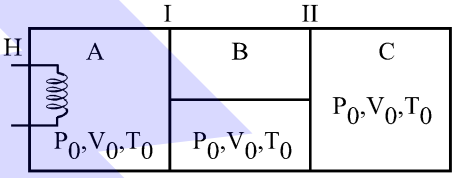
23. One mole of a gas mixture is heated under constant pressure, and heat required ΔQ is plotted against temperature difference acquired. Find the value of γ for mixture.



KT0011

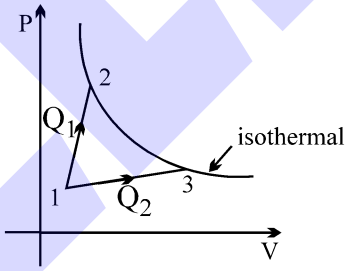
24. The figure shows an insulated cylinder divided into three parts A, B and C. Pistons I and II are connected by a rigid rod and can move without friction inside the cylinder. Piston I is perfectly conducting while piston II is perfectly insulating. The initial state of the gas ($\gamma = 1.5$) present in each compartment A, B and C is as shown. Now, compartment A is slowly given heat through a heater H such that the final volume of C becomes $\frac{4V_0}{9}$. Assume the gas to be ideal and find.

(a) Final pressures in each compartment A, B and C.
(b) Final temperatures in each compartment A, B and C
(c) Heat supplied by the heater
(d) Work done by gas in A and B.
(e) Heat flowing across piston I.



KT0021

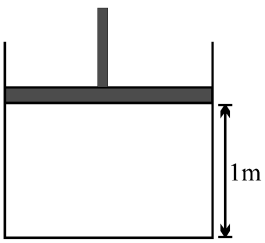
25. A gas takes part in two processes in which it is heated from the same initial state 1 to the same final temperature. The processes are shown on the P-V diagram by the straight line 1-2 and 1-3. 2 and 3 are the points on the same isothermal curve. Q_1 and Q_2 are the heat transfer along the two processes. Then in which case will the heat transfer be more.



KT0025

26. The piston cylinder arrangement shown contains a diatomic gas at temperature 300 K. The cross-sectional area of the cylinder is1 m^2. Initially the height of the piston above the base of the cylinder is 1 m. The temperature is now raised to 400 K at constant pressure. Find the new height of the piston above the base of the cylinder. If the piston is now brought back to its original height without any heat loss, find the new equilibrium temperature of the gas. You can leave the answer in fraction.

[JEE' 2004]



KT0005

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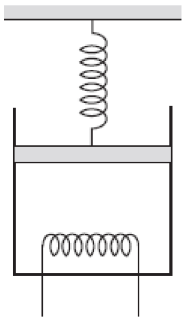
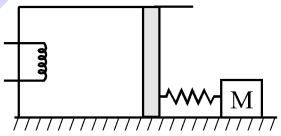
27. 2.00 mol of a monatomic ideal gas (U = 1.5 nRT) is enclosed in an adiabatic, fixed, vertical cylinder fitted with a smooth, light adiabatic piston. The piston is connected to a vertical spring of spring constant 200 N m^{-1} as shown in figure. The area of cross section of the cylinder is 20.0 cm^2. Initially, the spring is at its natural length and the temperature of the gas is 300 K. The atmospheric pressure is 100 kPa. The gas is heated slowly for some time by means of an electric heater so as to move the piston up through 10 cm. Find (a) the work done by the gas (b) the final temperature of the gas and (c) the heat supplied by the heater. **KT0382**



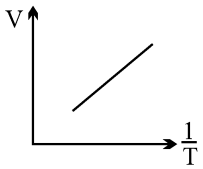
28. Two moles of an ideal monoatomic gas are confined within a cylinder by a massless & frictionless spring loaded piston of cross-sectional area 4 $\times 10^{-3}$ m^2. The spring is, initially in its relaxed state. Now the gas is heated by an electric heater, placed inside the cylinder, for some time. During this time, the gas expands and does 50 J of work in moving the piston through a distance 0.10 m. The temperature of the gas increases by 50 K. Calculate the spring constant & the heat supplied by the heater. [$P_0 = 10^5$ Pa] **KT0024**

29. An adiabatic cylinder has 8 gram of helium. A light smooth adiabatic piston is connected to a light spring of force constant 300 N/m. The other end of the spring is connected with a block of mass 1 kg kept on a rough horizontal surface of coefficient of friction $\mu = 0.3$. Area of cross-section of cylinder is A = 25 cm^2. Initially the spring is in a relaxed position and the temperature of the gas is 400 K. The gas is heated slowly for some time by means of an electric heater so as to bring the block M on the verge of motion. Take $P_{atm} = 10^5$ N/m^2. Find
(a) the work done by the gas
(b) the final temperature
(c) heat supplied by the heater



KT0028

30. One mole of an ideal monoatomic gas undergoes a process as shown in the figure. Find the molar specific heat of the gas in the process.



KT0014

31. Ideal diatomic gas is taken through a process $\Delta Q = 2\Delta U$. Find the molar heat capacity for the process (where ΔQ is the heat supplied and ΔU is change in internal energy) **KT0009**

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32. Two cylinders A and B of equal capacity are connected to each other via a stopcock. A contains a gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stopcock is suddenly opened. Answer the following :
(a) What is the final pressure of the gas in A and B ?
(b) What is the change in internal energy of the gas ?
(c) What is the change in the temperature of the gas ?
(d) Do the intermediate states of the system (before settling to the final equilibrium state) lie on its P-V-T surface ?

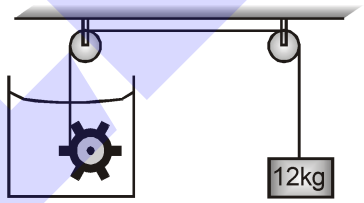
KT0015

33. A weightless piston divides a thermally insulated cylinder into two parts of volumes V and 3V. 2 moles of an ideal gas at pressure P = 2 atmosphere are confined to the part with volume V = 1 litre. The remainder of the cylinder is evacuated. The piston is now released and the gas expands to fill the entire space of the cylinder. The piston is then pressed back to the initial position. Find the increase of internal energy in the process and final temperature of the gas. The ratio of the specific heat of the gas $\gamma = 1.5$.

KT0023

34. A thermally insulated, closed copper vessel contains water at 15°C. When the vessel is shaken vigorously for 15 minutes, the temperature rises to 17°C. The mass of the vessel is 100 g and that of the water is 200 g. The specific heat capacities of copper and water are 420 J/kg-K and 4200 J/kg-K respectively. Neglect an thermal expansion.
[a] How much heat is transferred to the liquid-vessel system?
[b] How much work is done on the liquid-vessel system ?
[c] How much is the increase in internal energy of the system ? **KT0383**

35. Figure shows a paddle wheel coupled to a mass of 12 kg through fixed frictionless pulleys. The paddle is immersed in a liquid of heat capacity 4200 J/K kept in an adiabatic container. Consider a time interval in which the 12 kg block falls slowly through 70 cm.



[a] How much heat is given to the liquid ?
[b] How much work is done on the liquid ?
[c] Calculate the rise in the temperature of the liquid neglecting the heat capacity of the container and the paddle. **KT0384**

36. A 100 kg block is started with a speed of 2.0 m/s on a long, rough belt kept fixed in a horizontal position. The coefficient of kinetic friction between the block and the belt is 0.20.
[a] Calculate the chage in the internal energy of th block-belt system as the block comes to a stop on the belt.
[b] Consider the situation from a frame of reference moving at 2.0 m/s along the initial velocity of the block. As seen from this frame, the block is gently put on a moving belt an in due time the block starts moving with the belt at 2.0 m/s. Calculate the increase in the kinetic energy of the block as it stops slipping past the belt.
[c] Find the work done in this frame by the external force holding the belt. **KT0385**

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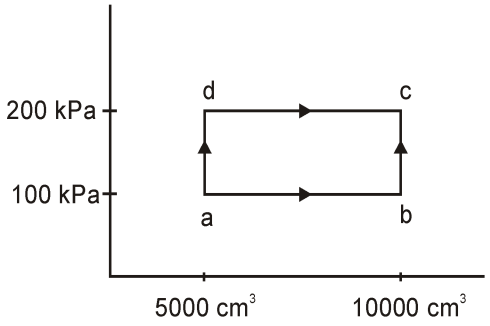
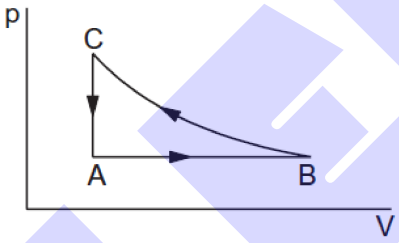
37. Half mole of ideal gas $\left(\gamma = \frac{5}{3}\right)$ is taken through the cycle abcda as shown in figure. Take $R = \frac{25}{3}$ J/mol-K.



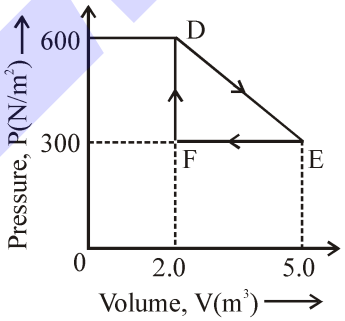
[a] Find the temperature of the gas in the states a, b, c and d.
[b] Find the amount of heat supplied in the process ab and bc.
[c] Find the amount of heat liberated in the process cd and da. **KT0386**

38. Consider the cyclic process ABCA on a sample of 2.0 mol of an ideal gas as shown in figure. The temperatures of the gas at A and B are 300 K and 500 K respectively. A total of 1200 J heat is withdrawn from the sample in the process. Find the work done by the gas in part BC. Take R = 8.3 JK^{-1} mol^{-1}.



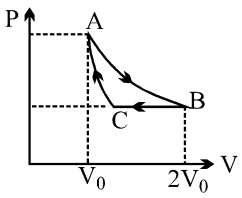
KT0387

39. A thermodynamic system is taken from an original state to an intermediate state by the linear process shown in figure. Its volume is then reduced to the original value from E to F by an isobaric process. Calculate the total work done by the gas from D to E to F



KT0012

40. In a cyclic process ABCA consisting of isothermal expansion AB, isobaric compression BC and adiabatic compression CA, find the efficiency of cycle (Given : $T_A = T_B = 400$ K, $\gamma = 1.5$)



KT0027

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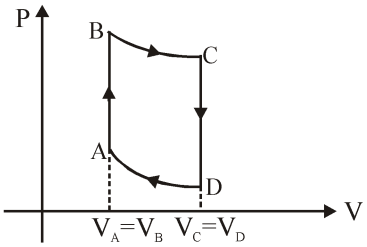
41. A cycle followed by an engine (made of one mole of perfect gas in a cylinder with a piston) is shown in figure.



A to B : volume constant

B to C : adiabatic

C to D : volume constant

D to A : adiabatic

$V_C=V_D= 2V_A =2V_B$

(a) In which part of the cycle heat is supplied to the engine from outside?

(b) In which part of the cycle heat is being given to the surrounding by the engine?

(c) What is the work done by the engine in one cycle? Write your answer in term of P_A, P_B, V_A

(d) What is the efficiency of the engine?

[$\gamma = \frac{5}{3}$ for the gas], ($C_V = \frac{3}{2}R$ for one mole) **KT0016**

42. A reversible heat engine has an efficiency of 33.3%, removing heat from a hot reservoir and rejecting heat to a cold reservoir at 0°C. If the engine now operates in reverse, how long would it take of freeze 1.0 kg of water at 0°C, if it operates on a power of 186 W? **KT0388**

43. Two engines A and B have their sources at 400 K and 350 K and sinks at 350 K and 300 K respectively. Which engine is more efficient and by how much ? **KT0389**

44. A carnot engine working between 400 K and 800 K has a work output of 1200 J per cycle. What is the amount of heat energy supplied to the engine from source per cycle? **KT0390**

45. A carnot engine works as a refrigrator between 250 K and 300 K. If it receives 750 cal of heat from the reservoir at the lower temperature. Calculate the amount of heat rejected at the higher temperature. **KT0391**

46. An ideal (Carnot) freezer in a kitchen has a constant temperature of 260 K, whereas the air in the kitchen has a constant temperature of 300 K. Suppose the insulation for the freezer is not perfect and conducts energy into the freezer at a rate of 0.150 W. Determine the average power required for the freezer's motor to maintain the constant temperature in the freezer. **KT0392**

47. A refrigerator takes heat from water at 0°C and transfer it to room at 27°C. If 100 kg of water is converted in ice at 0°C then calculate the workdone. (Latent heat of ice is 3.4×10^5 J/kg) **KT0393**

48. A heat engine employing a Carnot cycle with an efficiency of η = 10% is used as a refrigerating machine, the thermal reservoirs being the same. Find its coefficient of performance. **KT0394**

49. A steam turbine is operated with an intake temperature of 227°C and an exhaust temperature of 27°C. What is the maximum amount of work the turbine can do for a given heat input Q? Under what conditions is the maximum achieved? **KT0395**

50. A reversible heat engine receives heat inputs of 300 kJ and 200 kJ from two thermal reservoirs at 1000 K and 800 K, respectively. The engine rejects heat Q to a reservoir at 300 K. Find the value of Q and the work delivered by the engine. **KT0396**

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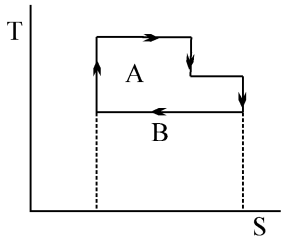
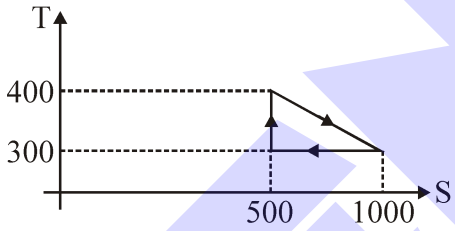
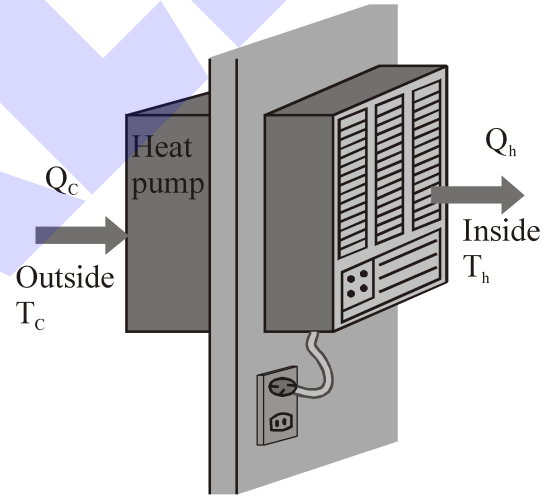
51. An engine is represented by a cyclic process shown in the T-S (temperature versus entropy)diagram. Here A represents the area of the region enclosed by the cycle and B of the area below the curve. Compute the efficiency cycle. **KT0397**



52. What is the efficiency for a reversible engine operating around the indicated cycle, where T is temperature in K and S is the entropy in joules/K ? **KT0398**



53. A heat pump used for heating shown in figure is essentially an air conditioner installed backward. It extracts energy from colder air outside and deposits it in a warmer room. Suppose the ratio of the actual energy entering the room to the work done by the device's motor is 20.0% of the theoretical maximum ratio. Determine the energy entering the room per joule of work done by the motor, given that the inside temperature is 27.0°C and the outside temperature is –3°C. **KT0399**



54. Find the change in entropy during the transition of 8 grammes of oxygen from a volume of 10 litres at a temperature of 80°C to a volume of 40 litres at a temperature of 300°C. **KT0400**

55. An ice tray contains 500 g of liquid water at 0°C. Calculate the change in entropy of the water as it freezes slowly and completely at 0°C. **KT0401**

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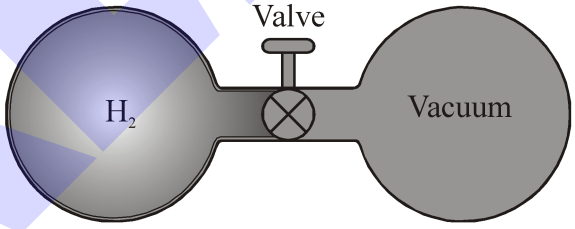
56. Find the entropy increment of one mole of carbon dioxide when its absolute temperature increases n = 2.0 times if the process of heating is : (C_v = 27 J/mol-K)
(a) isochoric ; (b) isbaric.
The gas is to be regarded as ideal. **KT0402**

57. Two moles of an ideal gas are cooled isochorically and then expanded isobarically to lower the gas temperature back to the initial value. Find the entropy increment of the gas if in this process the gas pressure changed n = 3.3 times. **KT0403**

58. A piece of copper of mass m_1 = 300 g with initial temperature t_1 = 97°C is placed into a calorimeter in which the water of mass m_2 = 100 g is at a temperature t_2 = 7°C. Find the entropy increment of the system by the moment the temperaturese equalize. The heat capacity of the calorimeter itself is negligibly small. (s_{cu} = 0.39 J/g-K) **KT0404**

59. (a) One kg of water at 0°C is brought into sudden contact with a large heat reservoir at 100°C. When the water has reached 100°C, what has been the change in entropy of the water? ofthe reservoir? Of the entire system consisting of both water and the heat reservoir?
(b) If the water had been heated from 0°C to 100°C by first bringing it into contact with a reservoir at 50°C and then another reservoir at 100°C, what would be the change in entropy of the entire system?
(c) Show how the water might be heated from 0°C to 100°C with negligible change in entropy of the entire system. **KT0405**

60. A 1.00 mol sample of H_2 gas is contained in the left side of the container shown in figure, which has equal volumes left and right. The right side is evacuated. When the valve is opened, the gas streams into the right side. What is the final entropy change of the gas? Assume the container is so large that the hydrogen behaves as an ideal gas. **KT0406**



61. A quantity of 0.20 mol of argon gas contained in a volume of 5.0 L mixes with 0.50 mol of neon gas contained in a volume of 5L, making a total volume of 10 L. Both gases are at the same temperature and can be regarded as ideal. Each volume is thermally isolated. What is the change in entropy ? **KT0407**

62. At very low temperatures the heat capacity fo crystals is equal to $C = aT^3$, where a is a constant. Find the entropy of a crystal as a function of temperature in this temperature interval. **KT0408**

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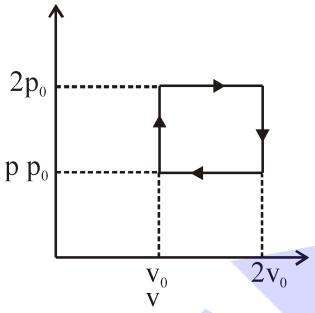
EXERCISE (JM)

1. The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monoatomic gas. The amount of heat, extracted from the source in a single cycle is :

[JEE-Mains-2013]



(1) p_0v_0 (2) $\left(\frac{13}{2}\right)p_0v_0$ (3) $\left(\frac{11}{2}\right)p_0v_0$ (4) $4p_0v_0$

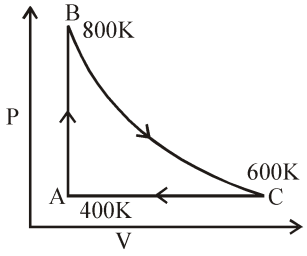
KT0109

2. An open glass tube is immersed in mercury in such a way that a length of 8 cm extends above the mercury level. The open of the tube is then closed and sealed and the tube is raised vertically up by addition 46 cm. What will be length of the air column above mercury in the tube now ?
(Atmospheric pressure = 76 cm of Hg) **[JEE-Mains-2014]**

(1) 38 cm (2) 6 cm (3) 16 cm (4) 22 cm

KT0110

3. One mole of diatomic ideal gas undergoes a cyclic process ABC as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are 400 K, 800 K and 600 K respectively. Choose the correct statement :



[JEE-Mains-2014]

(1) The change in internal energy in the process AB is –350 R.
(2) The change in internal energy in the process BC is –500R
(3) The change in internal energy in whole cyclic process is 250 R.
(4) The change in internal energy in the process CA is 700 R.

4. A solid body of constant heat capacity 1 J°/C is being heated by keeping it in contact with reservoirs in two ways - **[JEE-Mains-2015]**

(i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplies same amount of heat.
(ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat.

In both the cases body is brought from initial temperature 100°C to final temperature 200°C. Entropy change of the body in the two cases respectively is :

(1) ln2, 2ln2 (2) 2ln2, 8ln2 (3) ln2, 4ln2 (4) ln2, ln2 **KT0112**

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5. Consider a spherical shell of radius R at temperature T. The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $p = \frac{1}{3}\left(\frac{U}{V}\right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is :

[JEE-Mains-2015]

(1) $T \propto \frac{1}{R}$ (2) $T \propto \frac{1}{R^3}$ (3) $T \propto e^{-R}$ (4) $T \propto e^{-3R}$

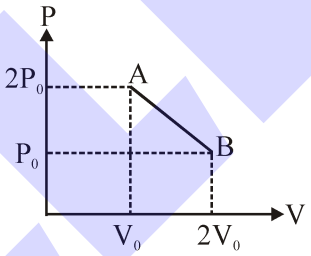
KT0113

6. Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^q, where V is the volume of the gas. The value of q is :- $\left(\gamma = \frac{C_p}{C_v}\right)$ **[JEE-Mains-2015]**

(1) $\frac{\gamma+1}{2}$ (2) $\frac{\gamma-1}{2}$ (3) $\frac{3\gamma+5}{6}$ (4) $\frac{3\gamma-5}{6}$

KT0114

7. 'n' moles of an ideal gas undergoes a process $A \rightarrow B$ as shown in the figure. The maximum temperature of the gas during the process will be : **[JEE-Mains-2016]**



(1) $\frac{9P_0V_0}{nR}$ (2) $\frac{9P_0V_0}{4nR}$ (3) $\frac{3P_0V_0}{2nR}$ (4) $\frac{9P_0V_0}{2nR}$

KT0115

8. An ideal gas undergoes a quasi static, reversible process in which its molar heat capacity C remains constant. If during this process the relation of pressure P and volume V is given by PV^n = constant, then n is given by (Here C_P and C_V are molar specific heat at constant pressure and constant volume, respectively) :- **[JEE-Mains-2016]**

(1) $n = \frac{C-C_V}{C-C_P}$ (2) $n = \frac{C_P}{C_V}$ (3) $n = \frac{C-C_P}{C-C_V}$ (4) $n = \frac{C_P-C}{C-C_V}$

KT0116

9. The temperature of an open room of volume 30 m^3 increases from 17°C to 27°C due to sunshine. The atmospheric pressure in the room remains 1×10^5 Pa. If n_i and n_f are the number of molecules in the room before and after heating, then $n_f - n_i$ will be : **[JEE-Main 2017]**

(1) 2.5×10^{25} (2) -2.5×10^{25} (3) -1.61×10^{23} (4) 1.38×10^{23}

KT0117

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\3-Calorimetry, Elasticity, T.E., H.T, KTG\Eng\02_KTG-TDY\01_Eng.p65

10. C_p and C_v are specific heats at constant pressure and constant volume respectively. It is observed that
$C_p - C_v = a$ for hydrogen gas
$C_p - C_v = b$ for nitrogen gas
The correct relation between a and b is : **[JEE-Main 2017]**

(1) a = 14 b (2) a = 28 b (3) $a = \frac{1}{14}b$ (4) a = b **KT0118**

11. Two moles of an ideal monoatomic gas occupies a volume V at 27°C. The gas expands adiabatically to a volume 2V. Calculate (a) the final temperature of the gas and (b) change in its internal energy.
[JEE-Main 2018]

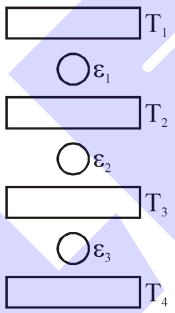
(1) (a) 195 K (b) –2.7 kJ (2) (a) 189 K (b) –2.7 kJ
(3) (a) 195 K (b) 2.7 kJ (4) (a) 189 K (b) 2.7 kJ **KT0119**

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

12. Two kg of a monoatomic gas is at a pressure of 4×10^4 N/m^2 . The density of the gas is 8 kg /m^3. What is the order of energy of the gas due to its thermal motion ? **[JEE-Main-2019_Jan]**

(1) 10^3 J (2) 10^5 J (3) 10^6 J (4) 10^4 J **KT0409**

13. Three Carnot engines operate in series between a heat source at a temperature T_1 and a heat sink at temperature T_4 (see figure). There are two other reservoirs at temperature T_2, and T_3, as shown, with $T_2 > T_2 > T_3 > T_4$. The three engines are equally efficient if: **[JEE-Main-2019_Jan]**



(1) $T_2 = \left(T_1^2 T_4\right)^{1/3}; T_3 = \left(T_1 T_4^2\right)^{1/3}$ (2) $T_2 = \left(T_1 T_4^2\right)^{1/3}; T_3 = \left(T_1^2 T_4\right)^{1/3}$

(3) $T_2 = \left(T_1^3 T_4\right)^{1/4}; T_3 = \left(T_1 T_4^3\right)^{1/4}$ (4) $T_2 = (T_1 T_4)^{1/2}; T_3 = (T_1^2 T_4)^{1/3}$ **KT0410**

14. In a process, temperature and volume of one mole of an ideal monoatomic gas are varied according to the relation VT = K, where K is a constant. In this process the temperature of the gas is incresed by ΔT. The amount of heat absorbed by gas is (R is gas constant) : **[JEE-Main-2019_Jan]**

(1) $\frac{1}{2}R\Delta T$ (2) $\frac{3}{2}R\Delta T$ (3) $\frac{1}{2}KR\Delta T$ (4) $\frac{2K}{3}\Delta T$ **KT0411**

15. An ideal gas is enclosed in a cylinder at pressure of 2 atm and temperature, 300 K. The mean time between two successive collisions is 6×10^{-8} s. If the pressure is doubled and temperature is increased to 500 K, the mean time between two successive collisions will be close to:
[JEE-Main-2019_Jan]

(1) 4×10^{-8}s (2) 3×10^{-6}s (3) 2×10^{-7}s (4) 0.5×10^{-8}s **KT0412**

16. The number density of molecules of a gas depends on their distance r from the origin as, $n(r) = n_0 e^{-\alpha r^4}$.
Then the total number of molecules is proportional to : **[JEE-Main-2019_April]**

(1) $n_0\alpha^{1/4}$ (2) $n_0\alpha^{-3}$ (3) $n_0\alpha^{-3/4}$ (4) $\sqrt{n_0}\,\alpha^{1/2}$ **KT0413**

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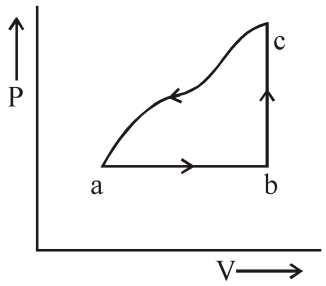
17. A Carnot engine has an efficiency of 1/6. When the temperature of the sink is reduced by 62ºC, its efficiency is doubled. The temperatures of the source and the sink are, respectively

[JEE-Main-2019_April]

(1) 124ºC, 62ºC (2) 37ºC, 99ºC (3) 62ºC, 124ºC (4) 99ºC, 37ºC **KT0414**

18. A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is –180J. The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is :

[JEE-Main-2019_April]



(1) 100 J (2) 120 J (3) 140 J (4) 130 J **KT0415**

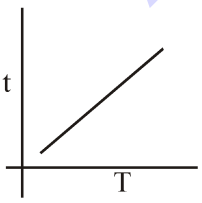
19. Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperatures, T_1 and T_2. The temperature of the hot reservoir of the first engine is T_1 and the temperature of the cold reservoir of the second engine is T_2. T is temperature of the sink of first engine which is also the source for the second engine. How is T related to T_1 and T_2, if both the engines perform equal amount of work ? **[JEE-Main-2020_Jan]**

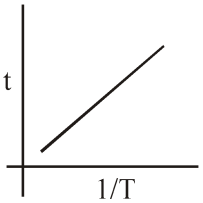
(1) $T = \frac{2T_1T_2}{T_1+T_2}$ (2) $T = \sqrt{T_1T_2}$ (3) $T = \frac{T_1+T_2}{2}$ (4) $T = 0$ **KT0416**

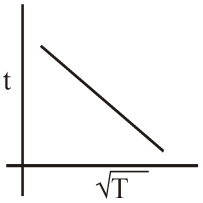
20. Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from τ_1 to τ_2. If $\frac{C_p}{C_v} = \gamma$ for this gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is given by : **[JEE-Main-2020_Jan]**

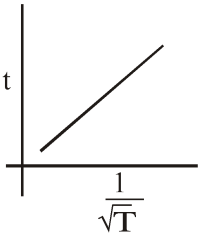
(1) $\left(\frac{1}{2}\right)^{\frac{\gamma+1}{2}}$ (2) 2 (3) $\frac{1}{2}$ (4) $\left(\frac{1}{2}\right)^{\gamma}$ **KT0417**

21. The plot that depicts the behavior of the mean free time t (time between two successive collisions) for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is: (Graphs are schematic and not drawn to scale) **[JEE-Main-2020_Jan]**

(1) 

(2) 

(3) 

(4) 

KT0418

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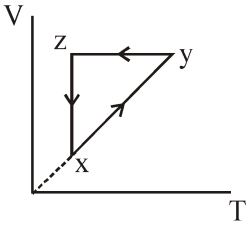
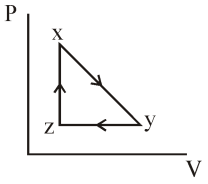
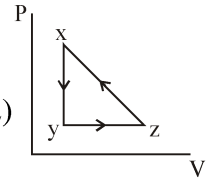
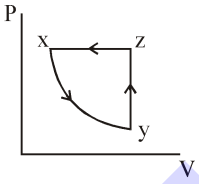
22. A thermodynamic cycle xyzx is shown on a V-T diagram. **[JEE-Main-2020_Jan]**

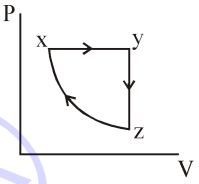


The P-V diagram that best describes this cycle is : (Diagrams are schematic and not to scale)

(1) 

(2) 

(3) 

(4) 

KT0419

23. Consider two ideal diatomic gases A and B at some temperature T. Molecules of the gas A are rigid, and have a mass m. Molecules of the gas B have an additional vibrational mode, and have a mass $\frac{m}{4}$. The ratio of the specific heats $(C_V^A$ and $C_V^B)$ of gas A and B, respectively is :

[JEE-Main-2020_Jan]

(1) 7 : 9 (2) 5 : 7 (3) 3 : 5 (4) 5 : 9 **KT0420**

24. An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true ? **[JEE-Main-2020_Sep]**

(A) the mean free path of the molecules decreases.

(B) the mean collision time between the molecules decreases.

(C) the mean free path remains unchanged.

(D) the mean collision time remains unchanged.

(1) (C) and (D) (2) (A) and (B) (3) (A) and (D) (4) (B) and (C) **KT0421**

25. The change in the magnitude of the volume of an ideal gas when a small additional pressure ΔP is applied at a constant temperature, is the same as the change when the temperature is reduced by a small quantity ΔT at constant pressure. The initial temperature and pressure of the gas were 300 K and 2 atm respectively. If $|\Delta T| = C|\Delta P|$ then value of C in (K/atm) is _____: **[JEE-Main-2020_Sep]**

KT0422

26. Number of molecules in a volume of 4 cm^3 of a perfect monoatomic gas at some temperature T and at a pressure of 2 cm of mercury is close to ? (Given, mean kinetic energy of a molecule (at T) is 4×10^{-14} erg, g = 980 cm/s^2, density of mercury = 13.6 g/cm^3) **[JEE-Main-2020_Sep]**

(1) 5.8×10^{18} (2) 5.8×10^{16} (3) 4.0×10^{18} (4) 4.0×10^{16} **KT0423**

27. On the basis of kinetic theory of gases, the gas exerts pressure because its molecules :

(1) continuously lose their energy till it reaches wall. **[JEE-Main-2021_Feb]**

(2) are attracted by the walls of container.

(3) continuously stick to the walls of container.

(4) suffer change in momentum when impinge on the walls of container. **KT0424**

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28. In a certain thermodynamical process, the pressure of a gas depends on its volume as kV^3. The work done when the temperature changes from 100°C to 300°C will be ___ nR, where n denotes number of moles of a gas. **[JEE-Main-2021_Feb]**

KT0425

29. A monoatomic gas of mass 4.0 u is kept in an insulated container. Container is moving with velocity 30 m/s. If container is suddenly stopped then change in temperature of the gas

(R = gas constant) is $\frac{x}{3R}$. Value of x is ___. **[JEE-Main-2021_Feb]**

KT0426

30. Given below are two statements :

Statement I : In a diatomic molecule, the rotational energy at a given temperature obeys Maxwell's distribution.

Statement II : In a diatomic molecule, the rotational energy at a given temperature equals the translational kinetic energy for each molecule.

In the light of the above statements, choose the correct answer from the options given below : **[JEE-Main-2021_Feb]**

(1) Statement I is false but Statement II is true.

(2) Both Statement I and Statement II are false.

(3) Both Statement I and Statement II are true.

(4) Statement I is true but Statement II is false. **KT0427**

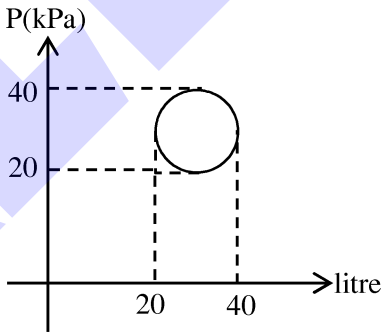
31. The internal energy (U), pressure (P) and volume (V) of an ideal gas are related as U = 3PV + 4. The gas is :- **[JEE-Main-2021_Feb]**

(1) Diatomic only (2) Polyatomic only

(3) Either monoatomic or diatomic (4) Monoatomic only **KT0428**

32. In the reported figure, heat energy absorbed by a system in going through a cyclic process is _____ πJ. **[JEE-Main-2021_July]**

KT0429



33. For a gas $C_P - C_V = R$ in a state P and $C_P - C_V = 1.10\ R$ in a state Q, T_P and T_Q are the temperatures in two different states P and Q respectively. Then **[JEE-Main-2021_July]**

(1) $T_P = T_Q$ (2) $T_P < T_Q$ (3) $T_P = 0.9\ T_Q$ (4) $T_P > T_Q$ **KT0430**

34. For an ideal gas the instantaneous change in pressure 'p' with volume 'v' is given by the equation $\frac{dp}{dv} = -ap$. If $p = p_0$ at $v = 0$ is the given boundary condition, then the maximum temperature one mole of gas can attain is : (Here R is the gas constant) **[JEE-Main-2021_Aug]**

(1) $\frac{p_0}{aeR}$ (2) $\frac{ap_0}{eR}$ (3) infinity (4) 0ºC **KT0431**

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35. An ideal gas is expanding such that PT^3 = constant. The coefficient of volume expansion of the gas is :
[JEE-Main-2021_Aug]

(1) $\frac{1}{T}$ (2) $\frac{2}{T}$ (3) $\frac{4}{T}$ (4) $\frac{3}{T}$ **KT0432**

36. A Carnot engine take 5000 kcal of heat from a reservoir at 727°C and gives heat to a sink at 127°C. The work done by the engine is :

(1) 3×10^6 J (2) Zero/शून्य (3) 12.6×10^6 J (4) 8.4×10^6 J
[JEE-Main-2022_June]
KT0433

37. A mixture of hydrogen and oxygen has volume 2000 cm^3, temperature 300 K, pressure 100 kPa and mass 0.76 g The ratio of number of moles of hydrogen to number of moles of oxygen in the mixture will be : **[JEE-Main-2022_June]**

(1) $\frac{1}{3}$ (2) $\frac{3}{1}$ (3) $\frac{1}{16}$ (4) $\frac{16}{1}$ **KT0434**

38. According to kinetic theory of gases, **[JEE-Main-2022_June]**

A. The motion of the gas molecules freezes at 0ºC

B. The mean free path of gas molecules decreases if the density of molecules is increased.

C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.

D. Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_BT$ (for monoatomic gases)

Choose the most appropriate answer from the options given below : **KT0435**

(1) A and C only (2) B and C only (3) A and B only (4) C and D only

39. Given below are two statement :

Statement - I : What μ amount of an ideal gas undergoes adiabatic change from state (P_1, V_1, T_1) to state (P_2, V_2, T_2), the work done is $W = \frac{1R(T_2 - T_1)}{1-\gamma}$, where $\gamma = \frac{C_P}{C_v}$ and R = universal gas constant,

Statement - II: In the above case. when work is done on the gas. the temperature of the gas would rise.

Choose the correct answer from the options given below: **[JEE-Main-2022_July]**

(1) Both statement-I and statement-II are true.

(2) Both statement-I and statement-II are false.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true. **KT0436**

40. A certain amount of gas of volume V at 27°C temperature and pressure 2×10^7 Nm^{-2} expands isothermally until its volume gets doubled. Later it expands adiabatically until its volume gets redoubled. The final pressure of the gas will be (Use $\gamma = 1.5$) **[JEE-Main-2022_July]**

(1) 3.536×10^5 Pa (2) 3.536×10^6 Pa (3) 1.25×10^6 Pa (4) 1.25×10^5 Pa
KT0437

41. A monoatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be:

(1) P (2) 8P (3) 32P (4) 64 P **KT0438**
[JEE-Main-2022_July]

42. One mole of a monoatomic gas is mixed with three moles of a diatomic gas. The molecular specific heat of mixture at constant volume is $\frac{\alpha^2}{4}R$ J/mol K; then the value of α will be _____. (Assume that the given diatomic gas has no vibrational mode.) **[JEE-Main-2022_July]**
KT0439

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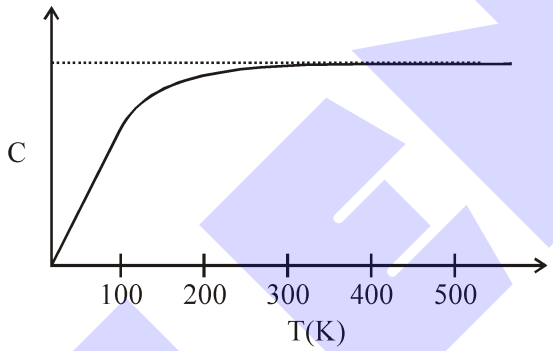
EXERCISE (JA)

1. Two non-reactive monoatomic ideal gases have their atomic masses in the ratio 2 : 3. The ratio of their partial pressures, when enclosed in a vessel kept at a constant temperature, is 4 : 3. The ratio of their densities is :- **[JEE-2013]**

(A) 1 : 4 (B) 1 : 2 (C) 6 : 9 (D) 8 : 9

KT0129

2. The figure below shows the variation of specific heat capacity (C) of a solid as a function of temperature (T). The temperature is increased continuously from 0 to 500 K at a constant rate. Ignoring any volume change, the following statement(s) is (are) correct to a reasonable approximation : **[JEE-2013]**



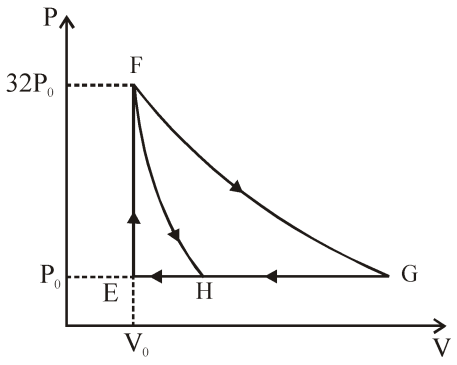
(A) the rate at which heat is absorbed in the range 0–100 K varies linearly with temperature T.

(B) heat absorbed in increasing the temperature from 0–100 K is less than the heat required for increasing the temperature from 400–500 K.

(C) there is no change in the rate of heat absorption in the range 400–500 K

(D) the rate of heat absorption increases in the range 200–300 K **KT0130**

3. One mole of a monatomic ideal gas is taken along two cyclic processes E→F→G→E and E→F→H→E as shown in the PV diagram. The processes involved are purely isochoric, isobaric, isothermal or adiabatic. **[JEE-2013]**



Match the paths in List I with the magnitudes of the work done in the List II and select the correct answer using the codes given blow the lists.

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List I		List II	
P.	G → E	1.	160 P_0V_0 ln2
Q.	G → H	2.	36 P_0V_0
R.	F → H	3.	24 P_0V_0
S.	F → G	4.	31 P_0V_0

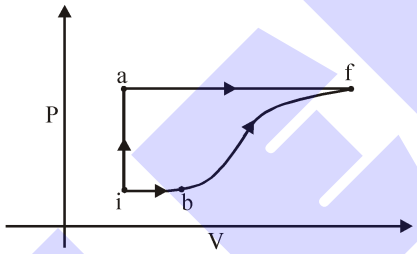
Codes :

	P	Q	R	S
(A)	4	3	2	1
(B)	4	3	1	2
(C)	3	1	2	4
(D)	1	3	2	4

KT0131

4. A thermodynamic system is taken from an initial state i with internal energy U_i = 100 J to the final state f along two different paths iaf and ibf, as schematically shown in the figure. The work done by the system along the paths af, ib and bf are W_{af} = 200 J, W_{ib} = 50 J and W_{bf} = 100 J respectively. The heat supplied to the system along the path iaf, ib and bf are Q_{iaf}, Q_{ib} and Q_{bf} respectively. If the internal energy of the system in the state b is U_b = 200 J and Q_{iaf} = 500 J, the ratio Q_{bf}/Q_{ib} is.

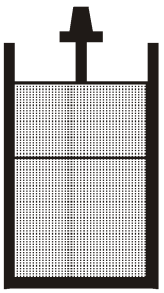
[JEE-Advance-2014]



KT0132

Paragraph for Questions 5 and 6

In the figure a container is shown to have a movable (without friction) piston on top. The container and the piston are all made of perfectly insulating material allowing no heat transfer between outside and inside the container. The container is divided into two compartments by a rigid partition made of a thermally conducting material that allows slow transfer of heat. The lower compartment of the container is filled with 2 moles of an ideal monatomic gas at 700 K and the upper compartment is filled with 2 moles of an ideal diatomic gas at 400 K. The heat capacities per mole of an ideal monatomic gas are $C_V = \frac{3}{2}R$, $C_P = \frac{5}{2}R$, and those for an ideal diatomic gas are $C_V = \frac{5}{2}R, C_P = \frac{7}{2}R$.



[JEE-Advance-2014]

5. Consider the partition to be rigidly fixed so that it does not move. When equilibrium is achieved, the final temperature of the gases will be

(A) 550 K (B) 525 K (C) 513 K (D) 490 K

KT0133

6. Now consider the partition to be free to move without friction so that the pressure of gases in both compartments is the same. Then total work done by the gases till the time they achieve equilibrium will be

(A) 250 R (B) 200 R (C) 100 R (D) –100 R **KT0133**

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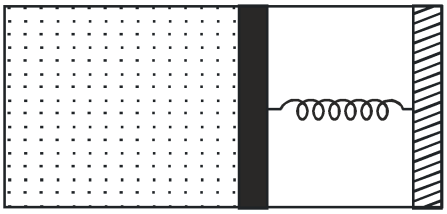
7. An ideal monoatomic gas is confined in a horizontal cylinder by a spring loaded piston (as shown in the figure). Initially the gas is at temperature T_1, pressure P_1 and volume V_1 and the spring is in its relaxed state. The gas is then heated very slowly to temperature T_2, pressure P_2 and volume V_2. During this process the piston moves out by a distance x. Ignoring the friction between the piston and the cylinder, the correct statement(s) is(are) :- **[JEE-Advance-2015]**



(A) If $V_2 = 2V_1$ and $T_2 = 3T_1$, then the energy stored in the spring is $\frac{1}{4}P_1V_1$

(B) If $V_2 = 2V_1$ and $T_2 = 3T_1$, then the change in internal energy is $3\ P_1V_1$

(C) If $V_2 = 3V_1$ and $T_2 = 4T_1$, then the work done by the gas is $\frac{7}{3}P_1V_1$

(D) If $V_2 = 3V_1$ and $T_2 = 4T_1$, then the heat supplied to the gas is $\frac{17}{6}P_1V_1$ **KT0135**

8. A gas is enclosed in a cylinder with a movable frictionless piston. Its initial thermodynamic state at pressure $P_i = 10^5$ Pa and volume $V_i = 10^{-3}$ m^3 changes to a final state at $P_f = \left(\frac{1}{32}\right) \times 10^5$ Pa and $V_f = 8 \times 10^{-3}$ m^3 in an adiabatic quasi-static process, such that P^3V^5 = constant. Consider another thermodynamic process that brings the system from the same initial state to the same final state in two steps: an isobaric expansion at P_i followed by an isochoric (isovolumetric) process at volumes V_f. The amount of heat supplied to the system in the two step process is approximately

[JEE-Advance 2016]

(A) 112 J (B) 294 J (C) 588 J (D) 813 J

KT0136

9. A flat plate is moving normal to its plane through a gas under the action of a constant force F. The gas is kept at a very low pressure. The speed of the plate v is much less than the average speed u of the gas molecules. Which of the following options is/are true ? **[JEE-Advance 2017]**

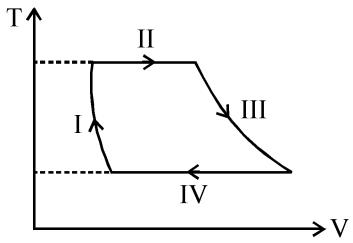
(A) The resistive force experienced by the plate is proportional to v

(B) The pressure difference between the leading and trailing faces of the plate is proportional to uv.

(C) The plate will continue to move with constant non-zero acceleration, at all times

(D) At a later time the external force F balances the resistive force. **KT0137**

10. One mole of a monatomic ideal gas undergoes a cyclic process as shown in the figure (whre V is the volume and T is the temperature). Which of the statements below is (are) true ? **[JEE-Advance 2018]**



(A) Process I is an isochoric process.

(B) In process II, gas absorbs heat.

(C) In process IV, gas releases heat.

(D) Processes I and II are not isobaric.

KT0139

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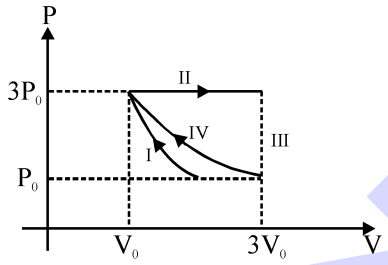
11. One mole of a monatomic ideal gas undergoes an adiabatic expansion in which its volume becomes eight times its initial value. If the initial temperature of the gas is 100 K and the universal gas constant R = 8.0 J mol^{-1} K^{-1}, the decrease in its internal energy, in Joule, is..... .

[JEE-Advance 2018]

KT0140

12. One mole of a monatomic ideal gas undergoes four thermodynamic processes as shown schematically in the PV-diagram below. Among these four processes, one is isobaric, one is isochoric, one is isothermal and one is adiabatic. Match the processes mentioned in List-I with the corresponding statements in List–II. **[JEE-Advance 2018]**



	List–I		List–II
P.	In process I	1.	Work done by the gas is zero
Q.	In process II	2.	Temperature of the gas remains unchanged
R.	In process III	3.	No heat is exchanged between the gas and its surroundings
S.	In process IV	4.	Work done by the gas is 6 P_0V_0

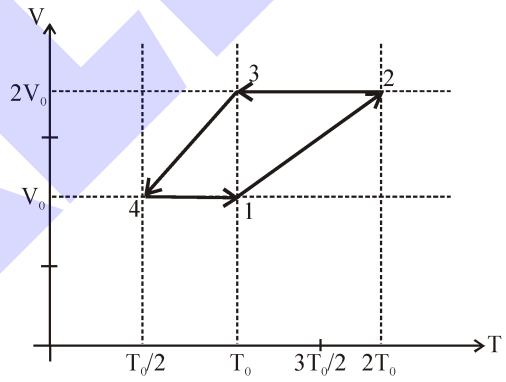
(A) P → 4 ; Q → 3 ; R → 1 ; S → 2 (B) P → 1 ; Q → 3 ; R → 2 ; S → 4

(C) P → 3 ; Q → 4 ; R → 1 ; S → 2 (D) P → 3 ; Q → 4 ; R → 2 ; S → 1

KT0141

13. One mole of a monoatomic ideal gas goes through a thermodynamic cycle, as shown in the volume versus temperature (V-T) diagram. The correct statement(s) is/are : [R is the gas constant]

[JEE-Advance 2019]



(A) Work done in this thermodynamic cycle (1→2→3→4→1) is $|W| = \frac{1}{2}RT_0$

(B) The ratio of heat transfer during processes 1 → 2 and 2 → 3 is $\left|\frac{Q_{1\to2}}{Q_{2\to3}}\right| = \frac{5}{3}$

(C) The above thermodynamic cycle exhibits only isochoric and adiabatic processes.

(D) The ratio of heat transfer during processes 1 → 2 and 3 → 4 is $\left|\frac{Q_{1\to2}}{Q_{3\to4}}\right| = \frac{1}{2}$ **KT0142**

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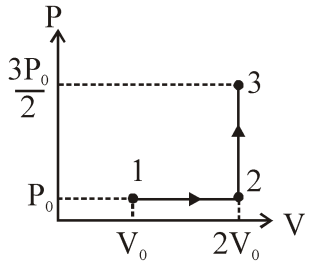
14. A mixture of ideal gas containing 5 moles of monatomic gas and 1 mole of rigid diatomic gas is initially at pressure P_0, volume V_0 and temperature T_0. If the gas mixture is adiabatically compressed to a volume $V_0/4$, then the correct statement(s) is/are,
(Give $2^{1.2} = 2.3$; $2^{3.2} = 9.2$; R is gas constant) **[JEE-Advance 2019]**
(A) The final pressure of the gas mixture after compression is in between $9P_0$ and $10P_0$
(B) The average kinetic energy of the gas mixture after compression is in between $18RT_0$ and $19RT_0$
(C) The work $|W|$ done during the process is $13RT_0$
(D) Adiabatic constant of the gas mixture is 1.6 **KT0143**

15. **Answer the following by appropriately matching the lists based on the information given in the paragraph.** **[JEE-Advance 2019]**
In a thermodynamics process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas $X = \frac{3}{2}R\ln\left(\frac{T}{T_A}\right) + R\ln\left(\frac{V}{V_A}\right)$. Here, R is gas constant, V is volume of gas, T_A and V_A are constants.
The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List-I	**List-II**
(I) Work done by the system in process $1 \to 2 \to 3$	(P) $\frac{1}{3}RT_0\ln 2$
(II) Change in internal energy in process $1 \to 2 \to 3$	(Q) $\frac{1}{3}RT_0$
(III) Heat absorbed by the system in process $1 \to 2 \to 3$	(R) RT_0
(IV) Heat absorbed by the system in process $1 \to 2$	(S) $\frac{4}{3}RT_0$
	(T) $\frac{1}{3}RT_0(3+\ln 2)$
	(U) $\frac{5}{6}RT_0$

If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is,



(A) I → Q, II → R, III → P, IV → U
(B) I → S, II → R, III → Q, IV → T
(C) I → Q, II → R, III → S, IV → U
(D) I → Q, II → S, III → R, IV → U
KT0144

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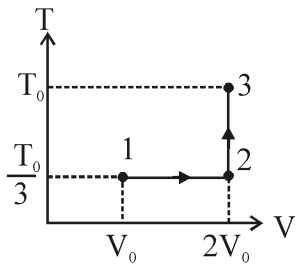
16. Answer the following by appropriately matching the lists based on the information given in the paragraph. **[JEE-Advance 2019]**

In a thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas $X = \frac{3}{2} R \ln\left(\frac{T}{T_A}\right) + R \ln\left(\frac{V}{V_A}\right)$. Here, R is gas constant, V is volume of gas, T_A and V_A are constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List-I	**List-II**
(I) Work done by the system in process $1 \to 2 \to 3$	(P) $\frac{1}{3}RT_0 \ln 2$
(II) Change in internal energy in process $1 \to 2 \to 3$	(Q) $\frac{1}{3}RT_0$
(III) Heat absorbed by the system in process $1 \to 2 \to 3$	(R) RT_0
(IV) Heat absorbed by the system in process $1 \to 2$	(S) $\frac{4}{3}RT_0$
	(T) $\frac{1}{3}RT_0(3 + \ln 2)$
	(U) $\frac{5}{6}RT_0$

If the process on one mole of monatomic ideal gas is an shown is as shown in the TV-diagram with $P_0V_0 = \frac{1}{3} RT_0$, the correct match is



(A) $I \to S, II \to T, III \to Q, IV \to U$
(B) $I \to P, II \to R, III \to T, IV \to S$
(C) $I \to P, II \to T, III \to Q, IV \to T$
(D) $I \to P, II \to R, III \to T, IV \to P$

KT0145

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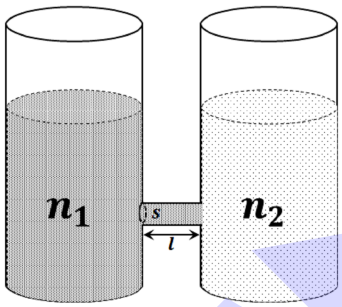
17. As shown schematically in the figure, two vessels contain water solutions (at temperature T) of potassium permanganate ($KMnO_4$) of different concentrations n_1 and n_2 ($n_1 > n_2$) molecules per unit volume with $\Delta n = (n_1 - n_2) << n_1$. When they are connected by a tube of small length ℓ and cross-sectional area S, $KMnO_4$ starts to diffuse from the left to the right vessel through the tube. Consider the collection of molecules to behave as dilute ideal gases and the difference in their partial pressure in the two vessels causing the diffusion. The speed v of the molecules is limited by the viscous force $-\beta v$ on each molecule, where β is a constant. Neglecting all terms of the order $(\Delta n)^2$, which of the following is/are correct? (k_B is the Boltzmann constant). **[JEE-Advance 2020]**



(A) the force causing the molecules to move across the tube is $\Delta n k_B TS$

(B) force balance implies $n_1 \beta v \ell = \Delta n k_B T$

(C) total number of molecules going across the tube per sec is $\left(\frac{\Delta n}{\ell}\right)\left(\frac{k_B T}{\beta}\right)S$

(D) rate of molecules getting transferred through the tube does not change with time **KT0440**

18. Consider one mole of helium gas enclosed in a container at initial pressure P_1 and volume V_1. It expands isothermally to volume $4V_1$. After this, the gas expands adiabatically and its volume becomes $32V_1$. The work done by the gas during isothermal and adiabatic expansion processes are W_{iso} and W_{adia}, respectively. If the ratio $\frac{W_{iso}}{W_{adia}} = f \ln 2$, then f is _____. **[JEE-Advance 2020]**

KT0441

19. A thermally isolated cylindrical closed vessel of height 8 m is kept vertically. It is divided into two equal parts by a diathermic (perfect thermal conductor) frictionless partition of mass 8.3 kg. Thus the partition is held initially at a distance of 4 m from the top, as shown in the schematic figure below. Each of the two parts of the vessel contains 0.1 mole of an ideal gas at temperature 300 K. The partition is now released and moves without any gas leaking from one part of the vessel to the other. When equilibrium is reached, the distance of the partition from the top (in m) will be _____ (take the acceleration due to gravity =10 ms^{-2} and the universal gas constant = 8.3 J $mol^{-1}K^{-1}$).

[JEE-Advance 2020]

KT0442

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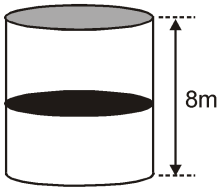
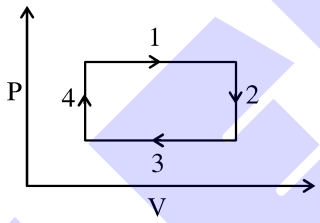


20. A spherical bubble inside water has radius R. Take the pressure inside the bubble and the water pressure to be p_0. The bubble now gets compressed radially in an adiabatic manner so that its radius becomes (R – a). For a << R the magnitude of the work done in the process is given by $(4\pi p_0 Ra^2)X$, where X is a constant and $\gamma = \frac{C_P}{C_V} = \frac{41}{30}$. The value of X is_____. **[JEE-Advance 2020]**

KT0443

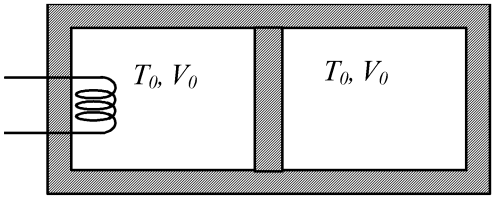
21. An ideal gas undergoes a four step cycle as shown in the P - V diagram below. During this cycle, heat is absorbed by the gas in **[JEE-Advance 2021]**



(A) steps 1 and 2 (B) steps 1 and 3
(C) steps 1 and 4 (D) steps 2 and 4 **KT0444**

Paragraph for Questions 22 and 23

A thermally insulating cylinder has a thermally insulating and frictionless movable partition in the middle, as shown in the figure below. On each side of the partition, there is one mole of an ideal gas, with specific heat at constant volume, $C_v = 2R$. Here, R is the gas constant. Initially, each side has a volume V_0 and temperature T_0. The left side has an electric heater, which is turned on at very low power to transfer heat Q to the gas on the left side. As a result the partition moves slowly towards the right reducing the right side volume to $V_0/2$. Consequently, the gas temperatures on the left and the right sides become T_L and T_R, respectively. Ignore the changes in the temperatures of the cylinder, heater and the partition. **[JEE-Advance 2021]**



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22. The value of $\frac{T_R}{T_0}$ is

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) 3 **KT0445**

23. The value of $\frac{Q}{RT_0}$ is

(A) $4\left(2\sqrt{2}+1\right)$ (B) $4\left(2\sqrt{2}-1\right)$ (C) $\left(5\sqrt{2}+1\right)$ (D) $\left(5\sqrt{2}-1\right)$ **KT0446**

24. List–I describes thermodynamic processes in four different systems. List–II gives the magnitudes (either exactly or as a close approximation) of possible changes in the internal energy of the system due to the process. **[JEE-Advance-2022]**

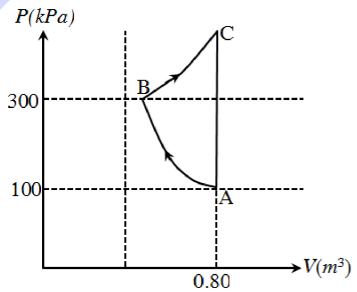
	List-I		**List-II**
(I)	10^{-3} kg of water at 100°C is converted to steam at the same temperature, at a pressure of 10^5 Pa. The volume of the system changes from 10^{-6} m^3 to 10^{-3} m^3 in the process. Latent heat of water = 2250 kJ/kg.	(P)	2 kJ
(II)	0.2 moles of a rigid diatomic ideal gas with volume V at temperature 500 K undergoes an isobaric expansion to volume 3 V. Assume R = 8.0 J $mol^{-1}K^{-1}$.	(Q)	7 kJ
(III)	On mole of a monatomic ideal gas is compressed adiabatically from volume $V=\frac{1}{3}m^3$ and pressure 2 kPa to volume $\frac{v}{8}$	(R)	4 kJ
(IV)	Three moles of a diatomic ideal gas whose molecules can vibrate, is given 9 kJ of heat and undergoes isobaric expansion.	(S) (T)	5 kJ 3 kJ

Which one of the following options is correct ?

(A) I → T, II → R, III → S, IV → Q (B) I → S, II → P, III → T, IV → P

(C) I → P, II → R, III → T, IV → Q (D) I → Q, II → R, III → S, IV → T **KT0447**

25. In the given P-V diagram, a monoatomic gas $\left(\gamma=\frac{5}{3}\right)$ is first compressed adiabatically from state A to state B. Then it expands isothermally from state B to state C. [Given: $\left(\frac{1}{3}\right)^{0.6}=0.5$, $\ln 2=0.7$].



Which of the following statement(s) is(are) correct? **[JEE-Advance-2022]**

(A) The magnitude of the total work done in the process A → B → C is 144 kJ.

(B) The magnitude of the work done in the process B → C is 84 kJ.

(C) The magnitude of the work done in the process A → B is 60 kJ.

(D) The magnitude of the work done in the process C → A is zero. **KT0448**

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	B	D	C	A	C	B	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	B	C	C	A	C	D	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	A	B	D	A	B	A	C	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	B	C	D	D	D	D	D	C	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	B	C	C	C	D	B	B	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	B	D	C	B	D	A	A	B	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	B	D	C	C	B	B	C	A	A	D
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	B	D	A	A	C	A	B	A	C	A
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	C	C	B	A	B	D	D	D	A	A
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	A	A	C	D	B	A	B	A	B	A
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	C	D	B	A	C	B,C	A,C	A,C,D	A,B,C	A,D
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	B,D	C,D	A,B,C,D	B,C	A,C	D	D	C,D	B	A

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	D	C	B	A	B	C	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	A	C	D	B	D	A	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	B	A	A	D	C	B	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	A	C,D	B,D	A,B	A,B	B,C	A,B,D	B	A
Que.	41	42	43	44	45					
Ans.	B	C	D	A-P,R; B-Q,S; C-Q,S; D-P,R	A-P; B-Q; C-Q					

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EXERCISE # S

1. 1.25×10^4 N/m^2 **2.** 327 °C **3.** (i) $P_1 < P_2$, $T_1 < T_2$; (ii) $T_1 = T_2 < T_3$; (iii) $V_2 > V_1$; (iv) $P_1 > P_2$

4. 36.5 cm **5. (i)** 74 cm; **(ii)** 73.94 cm, **(iii)** 69.52 cm **6.** 1 cm^{-3} ; 1 cm; 8.8×10^{35} m

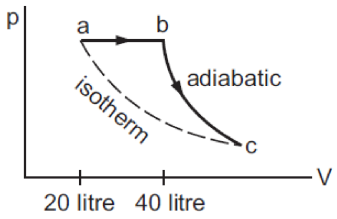
7. 6.3×10^{12} rad/sec **8.** $m = (1 - e^{-Mgh/RT})\, p_0 S/g$ **9.** $dT/dh = -Mg/R = -33$ mK/m.

10. (a) 1780 m/s; **(b)** 1.2×10^{28} **12. (a)** $C_V = 7/2R$; **(b)** $C_V = 19R/2$; **(c)** $C_V = 9R$ **13.** 3600 R

14. $\Delta T = \dfrac{mv_0^2}{3R}$ **15. 12600 J** **16. 40** **17.** 16.9 J

18. (i) $T_A = 120.28$ K, $T_B = 240.56$ K, $T_C = 481.12$ K, $T_D = 240.56$ K, **(ii)** No;
(iii) $\Delta Q_{ABC} = 3.25 \times 10^6$ J; $\Delta Q_{ADC} = 2.75 \times 10^6$ J

19. (a) 600 K, 900 K; **(b)** 10 J; **(c)** 14.9 J, 24.9 J; **(d)** 29.8 J **20. 2.64**

21. (a) 

; **(b)** 113.1 Litre, pressure = 0.44×10^5 N/m^2 ; **(c)** 1500 R

22. 800 kPa, 600 K in both cases **23.** 1.5

24. (a) Final pressure in A = $\dfrac{27}{8}P_0$ = Final pressure in C, Final pressure in B = $\dfrac{21}{4}P_0$

(b) Final temperature in A (and B) = $\dfrac{21}{4}T_0$, Final temperature in C = $\dfrac{3}{2}T_0$, **(c)** $18\, P_0V_0$

(d) work done by gas in A = $+P_0V_0$, work done by gas in B = 0, **(e)** $\dfrac{17}{2}P_0V_0$

25. $Q_1 < Q_2$ **26.** (a) 4/3 m, (b) $T_3 = 400\left(\dfrac{4}{3}\right)^{0.4}$ K **27. (a)** 21 J ; **(b)** 331 K ; **(c)** 793 J

28. 2000 N/m, 1295 J **29. (a)** 2.515, **(b)** 404.95 K, **(c)** 125.98 **30.** R/2

31. 5R **32. (a)** 0.5 atm; **(b)** zero; **(c)** zero; **(d)** no **33. 400 J, $2T_0$**

34. (a) Zero **(b)** 1764 J; **(c)** 1764 J **35. (a)** Zero **(b)** 84 J **(c)** 0.02°C **36. (a)** 0 J; **(b)** 200 J; **(c)** 400 J

37. (a) 120 K, 240 K, 480 K, 240 K; **(b)** 1250 J, 1500 J; **(c)** 2500 J, 750 J **38.** –4520 J **39.** 450 J

40. $1 - \dfrac{3\left(1 - \dfrac{1}{2^{1/3}}\right)}{\ell n\, 2}$ **41. (a)** A to B; **(b)** C to D; **(c)** $W_{AB} = \dfrac{3}{2}\left(1 - \left(\dfrac{1}{2}\right)^{2/3}\right)(P_B - P_A)V_A$; **(d)** $\left[1 - \left(\dfrac{1}{2}\right)^{2/3}\right]$

42. 15 min **43.** 1.8% **44.** 2400 joule per cycle **45.** 900 cal **46.** $\dfrac{3}{130}$ W

47. 3.36×10^6 J **48. 9** **49. (a)** 0.4Q ; **(b)** reversible engine **50. 335 kJ**

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51. A / (A + B) **52.** $\frac{1}{7}$ **53. 2** **54.** 5.4 J/k **55.** –615 J/k

56. (a) 19 J/K.mol); **(b)** 25 J/(K.mol) **57. 20 J/K** **58. 4.4 J/K**

59. (a) (i) $1000\ell n\frac{373}{273}$cal / k **(ii)** $-\frac{10^5}{373}$cal / k **(iii)** 44 cal/k approx;

(b) (i) $1000\ell n\frac{373}{273}$cal / k **(ii)** $-\frac{5\times10^4}{323}-\frac{5\times10^4}{373}$cal / k **(iii) 23 cal/k approx.**

60. 5.76 J/k **61.** Approx. 4 J/K **62.** $S=\frac{aT^3}{3}$

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	2	Bonus	1	1	2	3	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	1	1	1	3	4	4	3	1
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	2	4	150	3	4	50	3600	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	2	100	4	1	3	3	2	2	1	2
Que.	41	42								
Ans.	3	3								

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A,B,C,D or B,C,D	A	2	D	D	A,B,C	C	A,B,D	B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	900 (899.95, 900.05)	C	A,B	A,C,D	C	D	A,B,C	1.77-1.79	Bonus	2.05
Que.	21	22	23	24	25					
Ans.	C	A	B	C	B,C,D					

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Important Notes

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Important Notes

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